University of Cape Town

Time Series

Portfolio

The Temporal Shift

Author:
Isabella Lethbridge

Student Number:
LTHISA001

September 4, 2026

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Overview

Overview

Welcome to my portfolio for Time Series Analysis. This document showcases my learning progression, statistical understanding, computational applications, and reflections across the course.

Portfolio Structure

Reflective Commentary

Learning Achievements

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Methodological Challenges

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Ethical and Reproducibility Considerations

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Future Areas for Development

[cite: 1]

AI Use Log

Date	Tool Used	Prompt / Task Description	Link to Conversation	Verification & Changes Made
YYYY-MM-DD	ChatGPT / Claude	Debugging R code for model fit	Link	Checked output against manual calculations[cite: 1]

Section 1

Foundations

Learning outcomes

1. Explain what a time series is and how it differs from cross-sectional data, and say why temporal dependence calls for its own methods.
2. Treat an observed series as one realisation of a stochastic process.
3. Explain what forecasting is, judge when a process is predictable, and keep forecasts, goals and plans separate.
4. Match a forecast horizon to the decision it supports, and read a forecast as a random variable rather than a single number.
5. Identify trend, seasonality, cycle and irregular variation in a plot, and describe the shapes each of them takes in practice.
6. Produce the four benchmark forecasts, and evaluate any competing model against them on data it has not seen.
7. Use the lag and lead operators, and the algebra built on them, to write models compactly.
8. Define weak stationarity, say which of its conditions a given series violates, and choose a transformation that restores it.
9. Define white noise, MA, AR, ARMA, ARIMA and SARIMA processes, derive their basic properties, and identify them from an ACF and a PACF.
10. Fit, diagnose and forecast these models in R, following the workflow of Box and Jenkins, and test residuals against the white noise standard.
11. Explain volatility clustering and fit ARCH and GARCH models to it

What is a time series?

Time series is a single measurement taken at multiple points across time. Technically we are modelling the same thing over and over again (this makes sense as to why there is sequential correlation).

Note

Time series model dependence through sequential observations.

Assumption

Equally spaced measurements.

Assumption

Autocorrelation across time.

Qualitative, Continuous and Discrete Series

Example: Continuous series

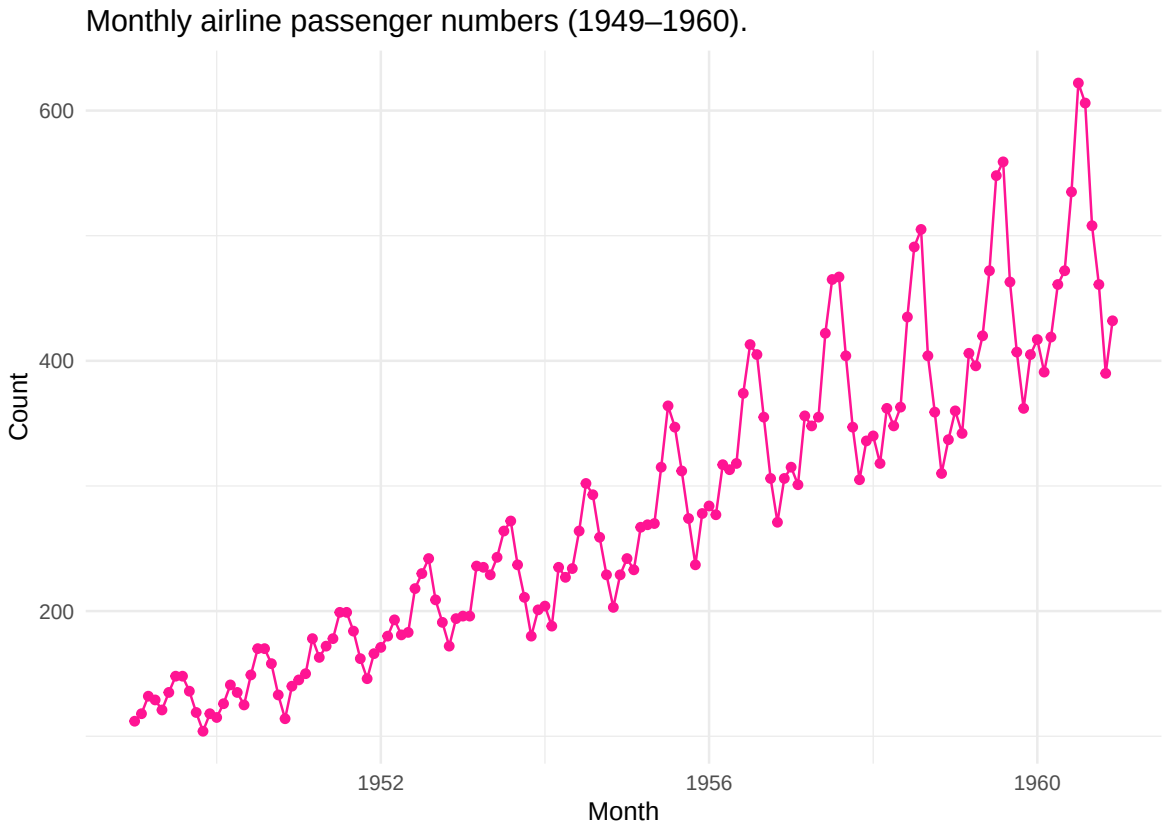


Figure 1: Time series plot of the number of monthly airline passengers from 1949 to 1960.

Figure 1 plots the number of monthly airline passengers over a 12 year period. This figure clearly illustrates the concepts of seasonality and trend.

Example: Discrete series

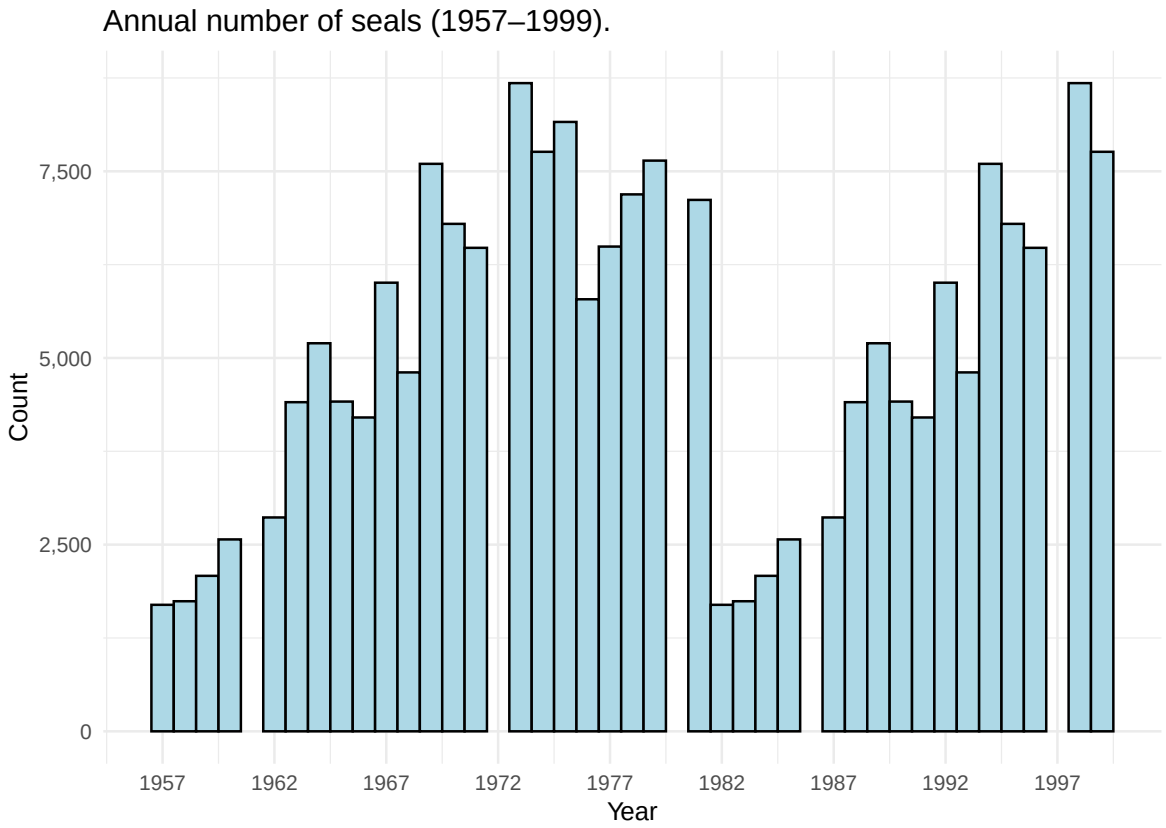


Figure 2: Time series plot of the number of seals from 1975 to 1999.

Figure 2 plots the changing number of observed seals over a period of 25 years.

What to do when you have missing observations in a time series?

In discrete time series to deal with missing observations you can express the model in state-space form (ARIMA or Structural Time Series). For a missing observation y_t, skip the update step in the kalman filter and only run the prediction step. The likelihood is evaluated only over the observed time points.

> **Kalman filtering**
>
> Kalman filtering (linear quadratic estimation) is an algorithm that uses a series of measurements over time to estimate unknown variables, by estimating a joint probability distribution over the variables for each time-step.

In continuous time series, missing observations look like irregularly spaced sampling intervals. You can model the underlying system using Stochastic Differential Equations (CARMA) or you can aggregate the irregular time steps into regular, discrete intervals.

Temporal granularity and the aggregation effect

Aggregating up can either smooth out high-frequency structure (stock returns, etc) or can reveal low-frequency structure (Sales, etc).

One realisation of a stochastic process

Noise in the data introduces different realisations. If your model fits one dataset perfectly then it is overfitting.

The Forecasting Problem

What is forecasting?

Forecasting is a prediction about what will happen in the future based on evidence that we have (with some level of uncertainty). This uncertainty gives rise to forecast intervals (not just point estimates). Goal setting is what we want to happen in the future. Planning helps us reach our goal.

> **Note**
>
> A series can only be called predictable if the interval and horizon are stated.

Forecast → Goal → Plan → Implementation → Outcome → Forecast …

A forecast horizon is how far into the future you try and predict values. When looking at time series data we are looking for stable, informative patterns (not just a lot of data).

A random walk is a process where memory is permanent. Every future shock accumulates endlessly resulting in forecast variance growing linearly. A stationary AR is a process where memory decays geometrically. Shocks dies out, forcing the forecast back to the process mean.

Stationary AR(1) process, h step forecast error variance:

$$\text{Var}(e_{T+h}) = \sigma^2 \frac{1-\phi^{2h}}{1-\phi^2} \to \frac{\sigma^2}{1-\phi^2} \quad \text{as } h \to \infty$$

> **Note**
>
> Forecasts are random variables.

Exploring a Series

The classical decomposition

$$\begin{aligned} \text{Additive:} \quad & y_t = T_t + S_t + C_t + I_t \\ \text{Multiplicative:} \quad & y_t = T_t \cdot S_t \cdot C_t \cdot I_t \end{aligned}$$

The nature of the seasonality in a series (additive, multiplicative or drifting) determines whether log transformation or differencing are required.

Two ways of looking at a series

1. Time domain - this looks at previous observations

You would use time domain when estimating predictive conditional expectations (forecasting at time $t+1$ given $t, t-1$), when dealing with short sample sizes.

A lag is the separation between two observations measured in time units.

> **Note**
>
> Plotting a series against its own first lag is the simplest possible diagnostic for dependence.

2. Frequency domain - this looks at previous trends

You would use frequency domain when identifying hidden cycles/patterns or when analysing long-memory dynamics.

The components of a series

Trend is the long run direction of movement. Seasonality is th pattern that repeats at a fixed and known interval. Cycles are long oscillations with no fixed length. Irregular variation (noise) is random variation unexplained by the former.

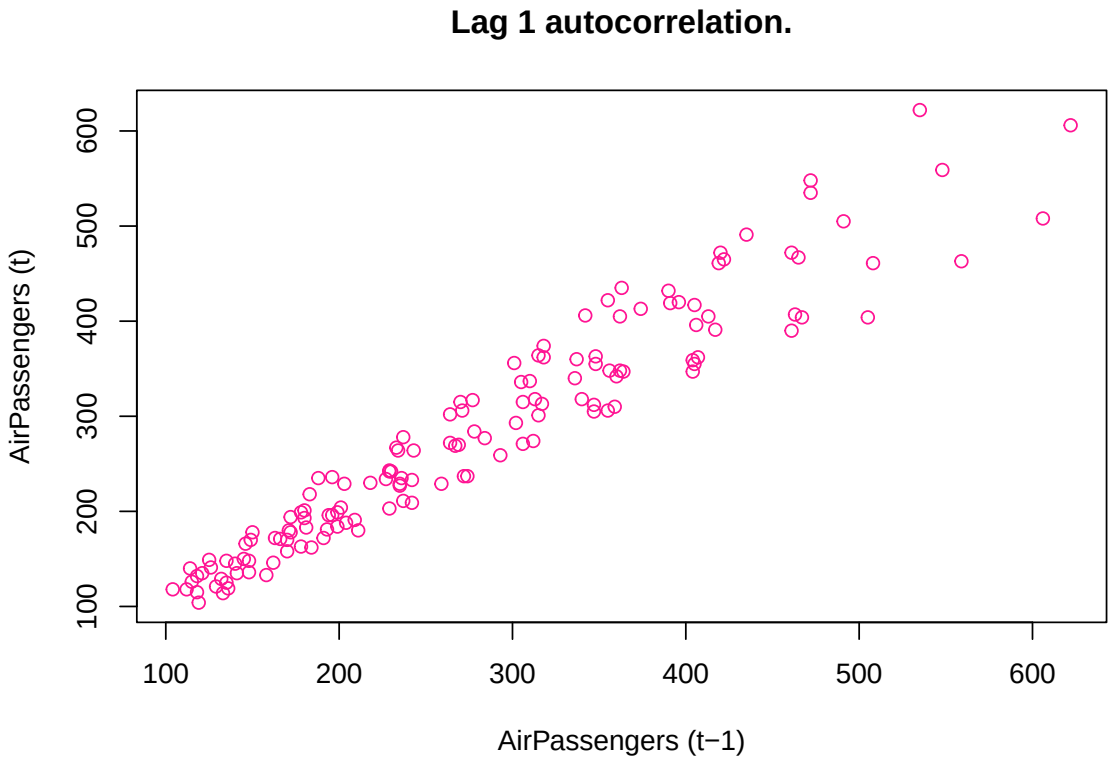


Figure 3: Simple diagnostic test for dependence. Autocorrelation of a series against its own first lag.

Time Series Analysis in R

`AirPassengers` dataset

```
# Run once - install anything missing
pkgs <- c("tidyverse", "tsibble", "feasts", "fable", "timetk",
          "modeltime", "kableExtra", "here")
install.packages(pkgs[!pkgs %in% installed.packages()[, "Package"]])
```

Handling dates and aggregation

`AirPassengers` is already a time series object but to convert a dataset to a time series object you use: `ts(c(), start = c(year, month), frequency = #obs/unit time)`

```
# Read in data:
data(AirPassengers)
# View(AirPassengers)

print(AirPassengers)
```

```
     Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1949 112 118 132 129 121 135 148 148 136 119 104 118
1950 115 126 141 135 125 149 170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
1952 171 180 193 181 183 218 230 242 209 191 172 194
1953 196 196 236 235 229 243 264 272 237 211 180 201
1954 204 188 235 227 234 264 302 293 259 229 203 229
1955 242 233 267 269 270 315 364 347 312 274 237 278
1956 284 277 317 313 318 374 413 405 355 306 271 306
1957 315 301 356 348 355 422 465 467 404 347 305 336
1958 340 318 362 348 363 435 491 505 404 359 310 337
1959 360 342 406 396 420 472 548 559 463 407 362 405
1960 417 391 419 461 472 535 622 606 508 461 390 432
```

```
# Sum by year:
annual_sum <- aggregate(AirPassengers, FUN = sum)

# Mean by year:
annual_mean <- aggregate(AirPassengers, FUN = mean)

print(annual_sum)
```

```
Time Series:
Start = 1949
End = 1960
Frequency = 1
 [1] 1520 1676 2042 2364 2700 2867 3408 3939 4421 4572 5140 5714
```

```
print(annual_mean)
```

```
Time Series:
Start = 1949
End = 1960
Frequency = 1
```

```
 [1] 126.6667 139.6667 170.1667 197.0000 225.0000 238.9167 284.0000
   328.2500
 [9] 368.4167 381.0000 428.3333 476.1667
```

Helper conversions

```r
t <- time(AirPassengers)

# Extract Year and Month
years <- floor(t)
months <- round(12 * (t - years)) + 1

head(data.frame(Decimal = as.numeric(t), Year = years, Month = months))
```

```
   Decimal Year Month
1 1949.000 1949     1
2 1949.083 1949     2
3 1949.167 1949     3
4 1949.250 1949     4
5 1949.333 1949     5
6 1949.417 1949     6
```

Subsetting with `window()`

```r
# Extract only the second half of 2023
sub_ts <- window(AirPassengers, start = c(1950, 7), end = c(1955, 10))
print(sub_ts)
```

```
     Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1950                         170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
1952 171 180 193 181 183 218 230 242 209 191 172 194
1953 196 196 236 235 229 243 264 272 237 211 180 201
1954 204 188 235 227 234 264 302 293 259 229 203 229
1955 242 233 267 269 270 315 364 347 312 274
```

Lags and Differences

```r
# Shift the time index forward by 1 period (does not move the values)
lagged_ts <- stats::lag(AirPassengers, k = -1)

# First-order differencing (removes linear trend)
diff_ts <- diff(AirPassengers, differences = 1)

# Seasonal differencing (removes a 12-month seasonal pattern)
seasonal_diff_ts <- diff(AirPassengers, lag = 12)

diff_ts
```

```
      Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
1949            6   14   -3   -8   14   13    0  -12  -17  -15   14
1950   -3   11   15   -6  -10   24   21    0  -12  -25  -19   26
```

```
1951     5     5    28   -15     9     6    21     0   -15   -22   -16    20
1952     5     9    13   -12     2    35    12    12   -33   -18   -19    22
1953     2     0    40    -1    -6    14    21     8   -35   -26   -31    21
1954     3   -16    47    -8     7    30    38    -9   -34   -30   -26    26
1955    13    -9    34     2     1    45    49   -17   -35   -38   -37    41
1956     6    -7    40    -4     5    56    39    -8   -50   -49   -35    35
1957     9   -14    55    -8     7    67    43     2   -63   -57   -42    31
1958     4   -22    44   -14    15    72    56    14  -101   -45   -49    27
1959    23   -18    64   -10    24    52    76    11   -96   -56   -45    43
1960    12   -26    28    42    11    63    87   -16   -98   -47   -71    42
```

Figure 3 illustrates the autocorrelation between `AirPassengers` and its lag 1.

Combing series

```
ts_a <- window(AirPassengers, start = c(1950, 1), end = c(1955, 12))
ts_b <- window(AirPassengers, start = c(1955, 1), end = c(1956, 12))

# Union: keeps full range, fills gaps with NA
combined_union <- ts.union(ts_a, ts_b)

# Intersect: keeps only the overlapping months
combined_intersect <- ts.intersect(ts_a, ts_b)

combined_intersect
```

```
         ts_a ts_b
Jan 1955  242  242
Feb 1955  233  233
Mar 1955  267  267
Apr 1955  269  269
May 1955  270  270
Jun 1955  315  315
Jul 1955  364  364
Aug 1955  347  347
Sep 1955  312  312
Oct 1955  274  274
Nov 1955  237  237
Dec 1955  278  278
```

Missing data

```
has_na <- anyNA(AirPassengers)
cat("Contains missing values:", has_na, "\n")
```

```
Contains missing values: FALSE
```

```
# Simple linear interpolation for isolated gaps (base R, no extra
    packages)
ts_with_gap <- AirPassengers
ts_with_gap[5] <- NA
ts_filled <- ts(approx(time(ts_with_gap),
                       ts_with_gap,
```

```
                        xout = time(ts_with_gap))$y,
                start = start(ts_with_gap),
                frequency = frequency(ts_with_gap))
ts_filled
```

```
     Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1949 112 118 132 129 132 135 148 148 136 119 104 118
1950 115 126 141 135 125 149 170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
1952 171 180 193 181 183 218 230 242 209 191 172 194
1953 196 196 236 235 229 243 264 272 237 211 180 201
1954 204 188 235 227 234 264 302 293 259 229 203 229
1955 242 233 267 269 270 315 364 347 312 274 237 278
1956 284 277 317 313 318 374 413 405 355 306 271 306
1957 315 301 356 348 355 422 465 467 404 347 305 336
1958 340 318 362 348 363 435 491 505 404 359 310 337
1959 360 342 406 396 420 472 548 559 463 407 362 405
1960 417 391 419 461 472 535 622 606 508 461 390 432
```

Time series plots

Figure 1 plots the `AirPassengers` time series.

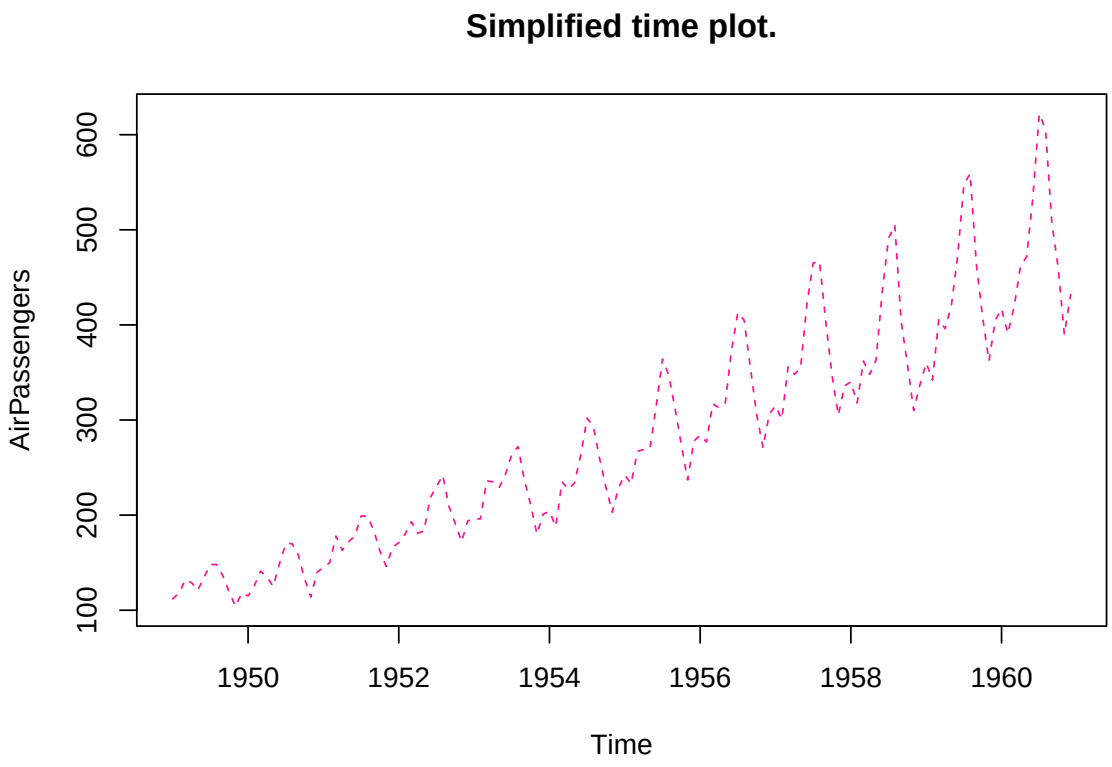


Figure 4: Time series plot of the number of monthly airline passengers from 1949 to 1960.

Check attributes of time series

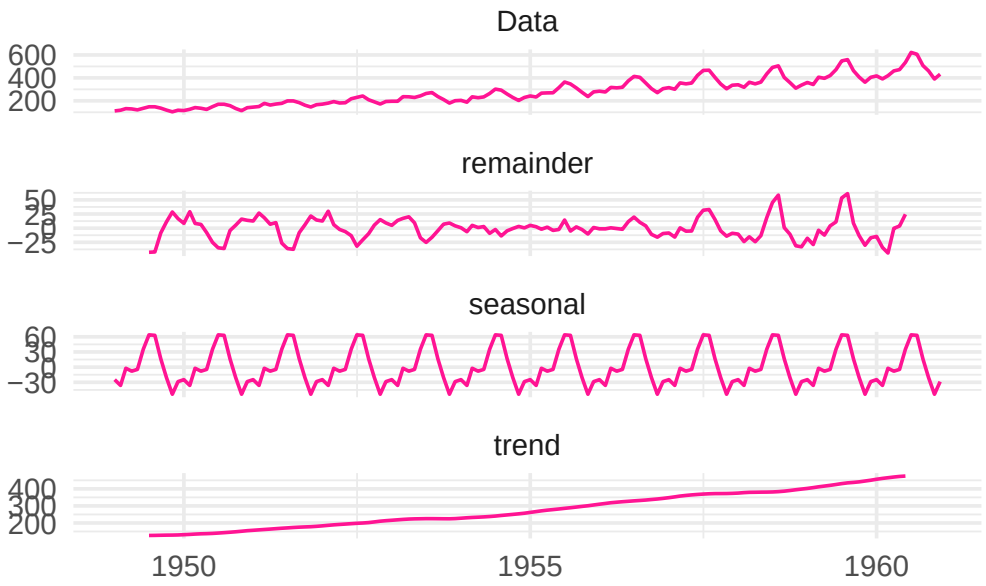
```
# Check the attributes
cat("Start:", start(AirPassengers), "\n")
```

```
Start: 1949 1
```

```
cat("Frequency:", frequency(AirPassengers), "\n")
```

```
Frequency: 12
```

```
library(ggfortify)
# A first look at structure: naive decomposition (covered in depth in
    Part B)
decomp <- decompose(AirPassengers)
autoplot(decomp,
     ts.colour='deeppink')+
  theme_minimal(base_size = 14)
```



Benchmarks and Forecast Accuracy

The benchmarks applied

Mean - the average of all previous observations is carried into the future.

$$\hat{y}_{T+h|T} = \bar{y}$$

Naive - the last observed value becomes the value that we take into the future.

$$\hat{y}_{T+h|T} = y_T$$

Seasonal naive - the last observed seasonal pattern is carried into the future.

$$\hat{y}_{T+h|T} = y_{T+h-m(k+1)}$$

Drift - The trend of the previsou observations is carried into the future.

$$\hat{y}_{T+h|T} = y_T + \frac{h(y_T - y_1)}{T-1}$$

> **Note**
>
> Residuals are for diagnostics (validation). Forecast errors are for evaluation (test).

Cross-validation

In time series successive observations are correlated and their ordering carries information about the data generating process. When performing model validation you cannot have future observations predicting the past.

Temporal leakage (look-ahead-bias) is when future observations enter model estimation, this artificially inflates accuracy.

The forecast origin is the last observation available when the forecast is made.

> **Note**
>
> The test set should be at least as long as the largest horizon of interest.

Strategies

1. Walk-forward validation

The forecast origin advances one step at a time.

2. Expanding window validation

Every historical observation is kept. The training set grows with all past data. When older data is more informative this method reduces estimation variance.

3. Sliding window validation

The training window has a fixed length. The number of training observations stays the same but the forecast origin is moved forward through time. When older data is less informative this method reduces bias.

What is the difference between interpolation and forecasting?

> **Note**
>
> Don't ignore structural breaks when choosing a validation strategy.

Manual cross-validation

Cross-validation is performed using an expanding window approach.

```r
# Load data:
data(AirPassengers)
dat = AirPassengers

# Load libraries:
library(forecast)

n = length(dat) # no. observations
train_n = 60    # no. training data points
h = 12          # forecast horizon

# Initialise error matrix:
names  = c('Mean','Naive','Seasonal naive', 'Drift')
splits = n-train_n-h+1

res = matrix(NA,
             nrow=(n-h)-train_n+1,
             ncol=4,
             dimnames=list(NULL,names))

for (i in train_n:(n-h)) {
  idx = i-train_n+1
  # Subset training data:
  train = window(dat,
                 end=c(1949+(i-1)%/%12,(i-1)%%12+1))

  # Extract test data:
  test = window(dat,
                start=c(1949+i%/%12,i%%12+1),
                end=c(1949+(i+h-1)%/%12,(i+h-1)%%12+1))

  # Fit model:
  fits = list(
    Mean   = meanf(train,h=h),
    Naive  = naive(train,h=h),
    SNaive = snaive(train,h=h),
    Drift  = rwf(train,h=h,drift=TRUE)
  )

  # Extract rmse for each model:
  rmse = sapply(fits,\(f) sqrt(mean((as.numeric(test)-as.numeric(f$mean))
    ^2)))

```

```
  # Store results:
  res[idx,] = rmse
}

colMeans(res)
```

```
          Mean          Naive Seasonal naive          Drift
     152.87587       74.08791       39.92124       71.96374
```

Cross-validation using `tsCV()`

```
tsCV_m = tsCV(AirPassengers,meanf,h=1,window=60)
m_err  = sqrt(mean(tsCV_m^2, na.rm = TRUE))

tsCV_n = tsCV(AirPassengers,naive,h=1,window=60)
n_err  = sqrt(mean(tsCV_n^2, na.rm = TRUE))

tsCV_sn = tsCV(AirPassengers,snaive,h=1,window=60)
sn_err  = sqrt(mean(tsCV_sn^2, na.rm = TRUE))

tsCV_d = tsCV(AirPassengers,rwf,h=1,window=60)
d_err  = sqrt(mean(tsCV_d^2, na.rm = TRUE))

err_df = cbind(m_err,n_err,sn_err,d_err)
colnames(err_df) = c('Mean','Naive','Seasonal naive','Drift')
err_df
```

```
         Mean    Naive Seasonal naive    Drift
[1,] 102.0794 41.52524       40.61316 41.52524
```

When comparing the expanding window vs the sliding window, the sliding window gives lower errors for all models except the seasonal naive model. This is because the `AirPassengers` dataset does not have constant variance, as seen in Figure 1.

Exercises

1. Why is ordinary random k-fold cross-validation inappropriate for forecasting?

If you exclude an observation at time t-1 but use observation t, you are technically predicting the past based on future observations. Future observations must never be used when estimating a forecasting model as this violates a fundamental principal of forecasting -

2. Define temporal leakage.

Future observations enter model estimation. Reported accuracy looks excellent but is unattainable in practice - the future is unknown when real forecasts are made.

3. Contrast Walk_Forward, Expanding Window and Sliding Window validation.

In expanding window validation every historical observation is kept. The training set grows with all past data. In sliding window validation the training window has a fixed

length. The number of training observations stays the same but the forecast origin is moved forward through time. Walk-forward validation is when the forecast origin advances one step at a time.

4. When would Sliding Window outperform Expanding Window?

When data is non-stationary and older data is less informative.

5. A firm refits its model daily using only the most recent 250 observations. Which strategy is this?

Sliding window

6. Why must temporal ordering be preserved during validation?

Otherwise the model knows things it could not have known at forecasting time. This leads to inflated accuracy that is unattainable in practise.

Operators, Stationarity and White Noise

Lag and lead operators

Lag operator $L_{y_t} = y_{t-1}$ at time t uses the observation recorded at time $t-1$.

> **Note**
>
> The subscript records where a value came from, not where it is displayed.

Lead operator $F_{y_t} = y_{t+1}$ at time t uses the observation recorded at time $t+1$, $F = L^{-1}$.

Lag and lead in R

```r
y <- c(3, 7, 4, 8, 5)
c(NA, y[1:(length(y)-1)]) # lag by hand
```

```
[1] NA  3  7  4  8
```

```r
dplyr::lag(y, n = 1) # same shift
```

```
[1] NA  3  7  4  8
```

```r
dplyr::lead(y, n = 1) # lead
```

```
[1]  7  4  8  5 NA
```

```r
lag_twice = dplyr::lag(dplyr::lag(y,n=1),n=1)
lag_twice
```

```
[1] NA NA  3  7  4
```

```r
dplyr::lag(y,n=2)
```

```
[1] NA NA  3  7  4
```

> ⚠ Built in `lag()` functions
>
> Don't get confused between `stats::lag()` and `dplyr::lag()`. They do different things!

Differencing

First difference:

$$\Delta y_t = y_t - y_{t-1} = (1 - L)y_t,$$

Seasonal difference (at lag 12):

$$\Delta_{12} y_t = y_t - y_{t-12} = (1 - L^{12})y_t,$$

```
# First differences, by hand and with diff().
y2 <- c(10, 6, 9, 12, 7)
y2 - dplyr::lag(y2, 1)
```

```
[1] NA -4  3  3 -5
```

```
diff(y2) # Note: no NA added
```

```
[1] -4  3  3 -5
```

Weak and strict stationarity

When we have a multiplicative model, with expanding variance, we can log-transform it to obtain an additive model. This removes the variance of seasonality but as seen below the series still has a trend.

Taking differences between observations removes the linear trend and taking seasonal differences removes the seasonal trend.

> ❗ Assumption
>
> No trends present in data.

We can remove both the linear and seasonal trend in the data:

```
# Check what sasonal and what regular differences are required:
nsdiffs(log_AP)
```

```
[1] 1
```

```
ndiffs(log_AP)
```

```
[1] 1
```

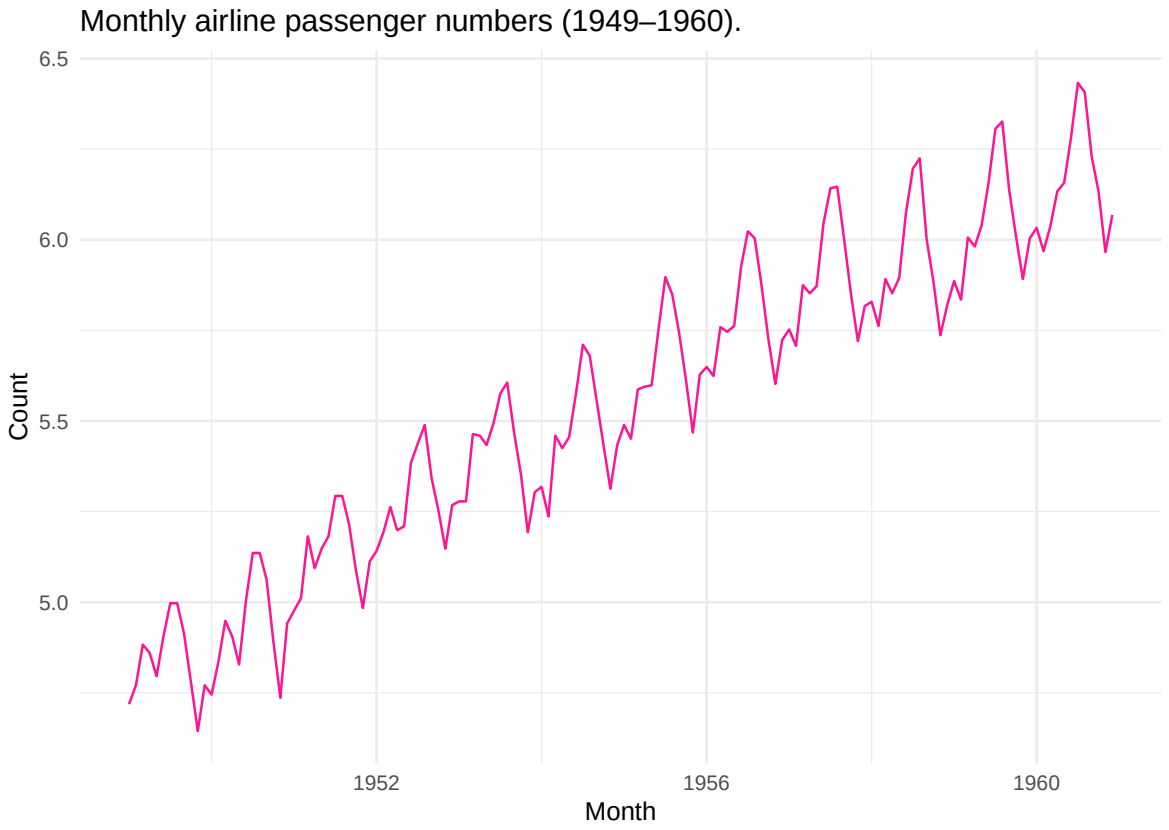


Figure 5: Time series plot of the log-tranformed number of monthly airline passengers from 1949 to 1960.

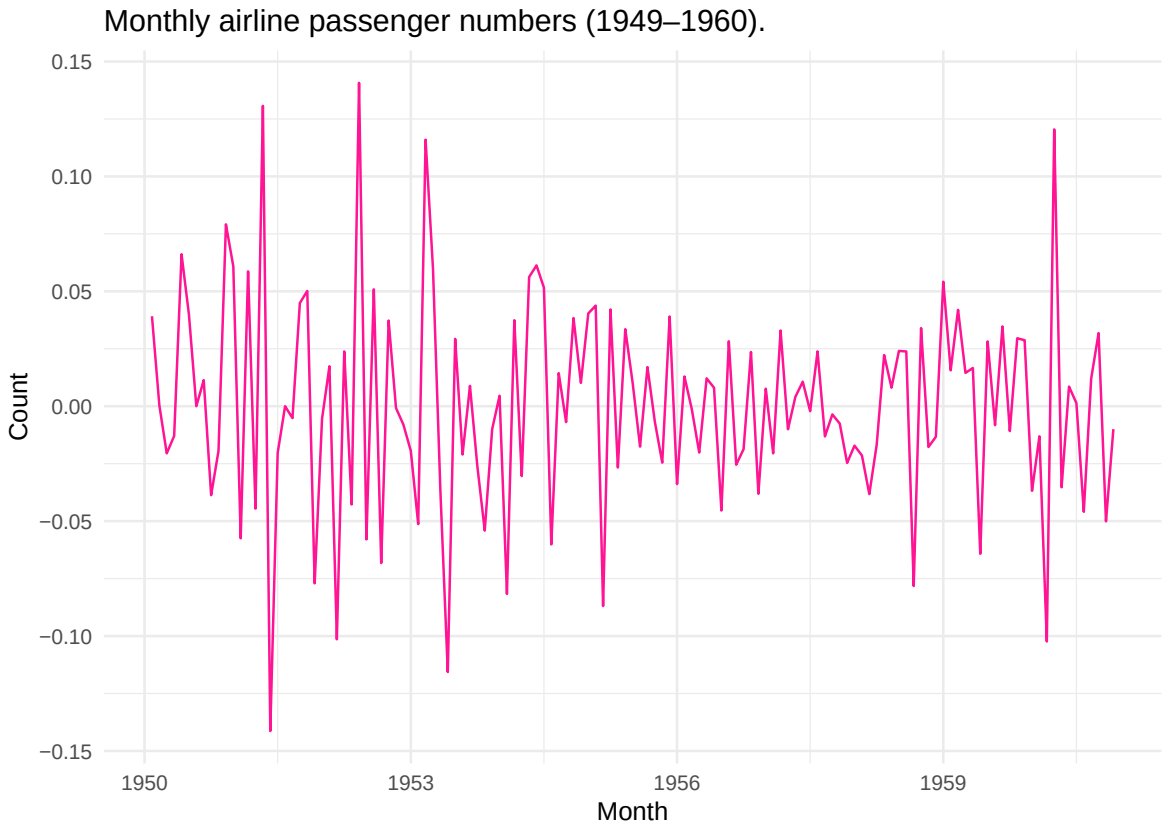


Figure 6: Time series plot of the number of monthly airline passengers from 1949 to 1960 in a stationary distribution.

White noise

Weak white noise only requires uncorrelated observations. Strict white noise requires *iid* observations.

White noise means unpredictible variance, whatever the variance.

> **Note**
>
> Indicator variables can be included to remove trends when the series has structural breaks in it.

Linear Models for the Conditional Mean

Although white noise is uncorrelated, a weighted combination of neighbouring white noise is not.

Filtering is the process of building a new series as a weighted combination of the terms of another.

The MA(q) model

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q)}, \quad \varepsilon_t \sim WN(0,\sigma^2).$$

The moving average model is driven by lagged shocks not lagged observations. A MA(q) process is stationary for any values of the coefficients, since it is a finite weighted sum of white noise terms with fixed weights. The coefficients must satisfy invertibility.

The AR(p) model

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

For the autoregressive model there is a set of correlation in the structure. The current value of a variable depends linearly on its own past. An AR(p) process models memory.

If you are observing spikes outside of the first few this indicates seasonality in the series. Figure 8 suggests that there is still seasonality present in the transformed and differenced `AirPassengers` dataset, however this could also just be because differencing reduced the number of observations resulting in a more noisy dataset.

The ARMA(q, q) model

The ARMA model allows for process that show persistence and short lived shocks simultaneously. The series must be stationary.

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

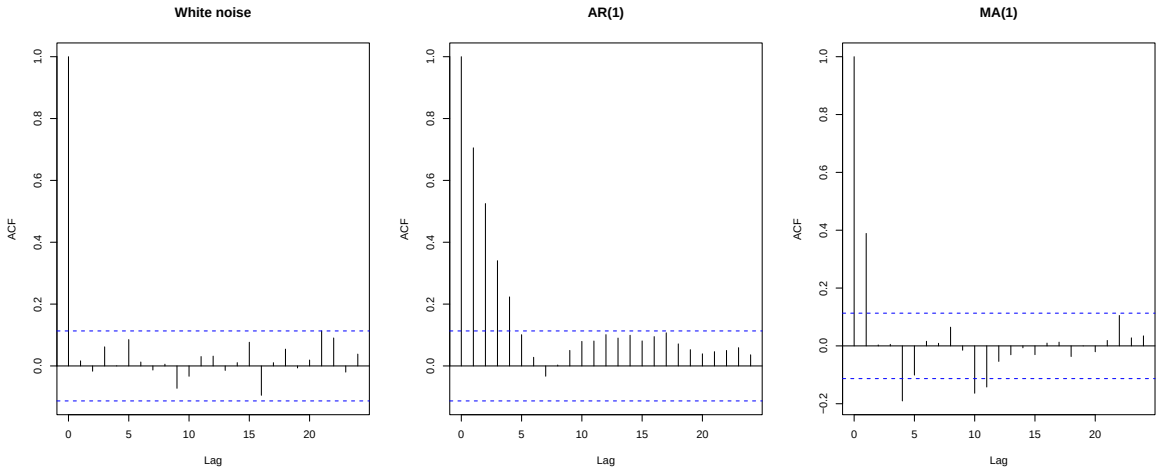


Figure 7: Three processes.

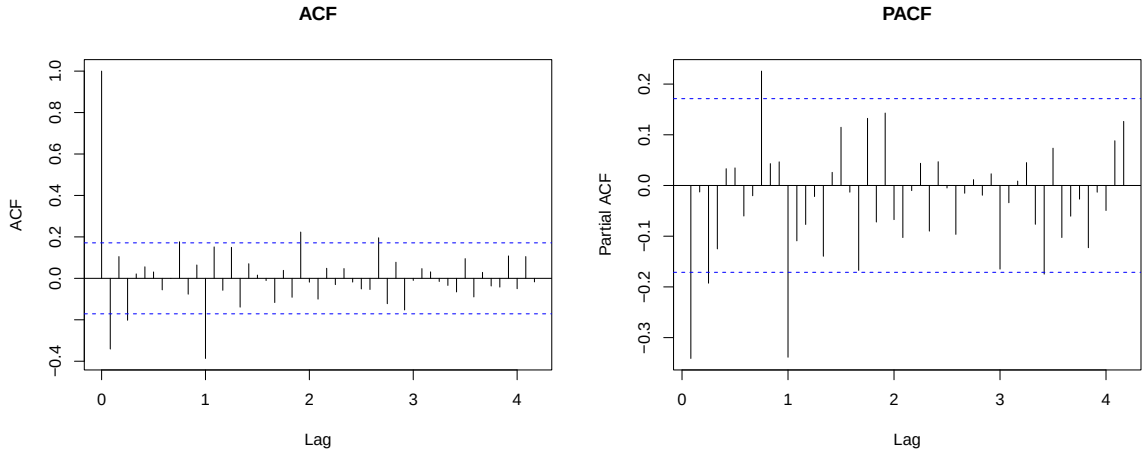


Figure 8: ACF and PACF plots for the AirPassengers dataset.

this is equivalent to: $\phi(L)y_t = c + \theta(L)\varepsilon_t$.

> **Note**
>
> ARMA models are fitted by maximum likelihood due to the moving average part (not OLS).

The ARIMA(p, d, q) model

The ARIMA model applies d difference to y_t and then fits an ARMA(p, q) to the result.

$$\phi(L)(1 - L)^d y_y = c + \theta(L)\varepsilon_t$$

Table 2: Special cases of ARIMA.

Specification	Equivalent to
ARIMA(1,0,0)	AR(1)
ARIMA(0,0,1)	MA(1)
ARIMA(1,0,1)	ARMA(1,1)
ARIMA(0,1,0)	Random walk
ARIMA(0,1,0) with c 0	Random walk with drift
ARIMA(1,1,1)	One difference, then ARMA(1,1)

Although Table 2 illustrates the special cases of ARIMA, sometimes there is a constant included that must be taken into account.

```

    Ljung-Box test

data:  Residuals from ARIMA(1,1,1)
Q* = 231.49, df = 22, p-value < 2.2e-16

Model df: 2.   Total lags used: 24
```

```
NULL
```

In Figure 10 we can see that the residuals do not look like white noise, this is because the seasonality hasn't been removed from the data with an ARIMA(1,1,1) model.

The SARIMA$(p, d, q)(P, D, Q)_m$ model

$$\Phi(L^m)\phi(L)(1 - L^m)^D(1 - L)^d y_t = \Theta(L^m)\theta(L)\varepsilon_t.$$

When we check an ACF plot of the `AirPassengers` series, we can see clear seasonality.

Figure 11 and Figure 10 suggest that we need to model the `AirPassengers` series with a seasonality component as well.

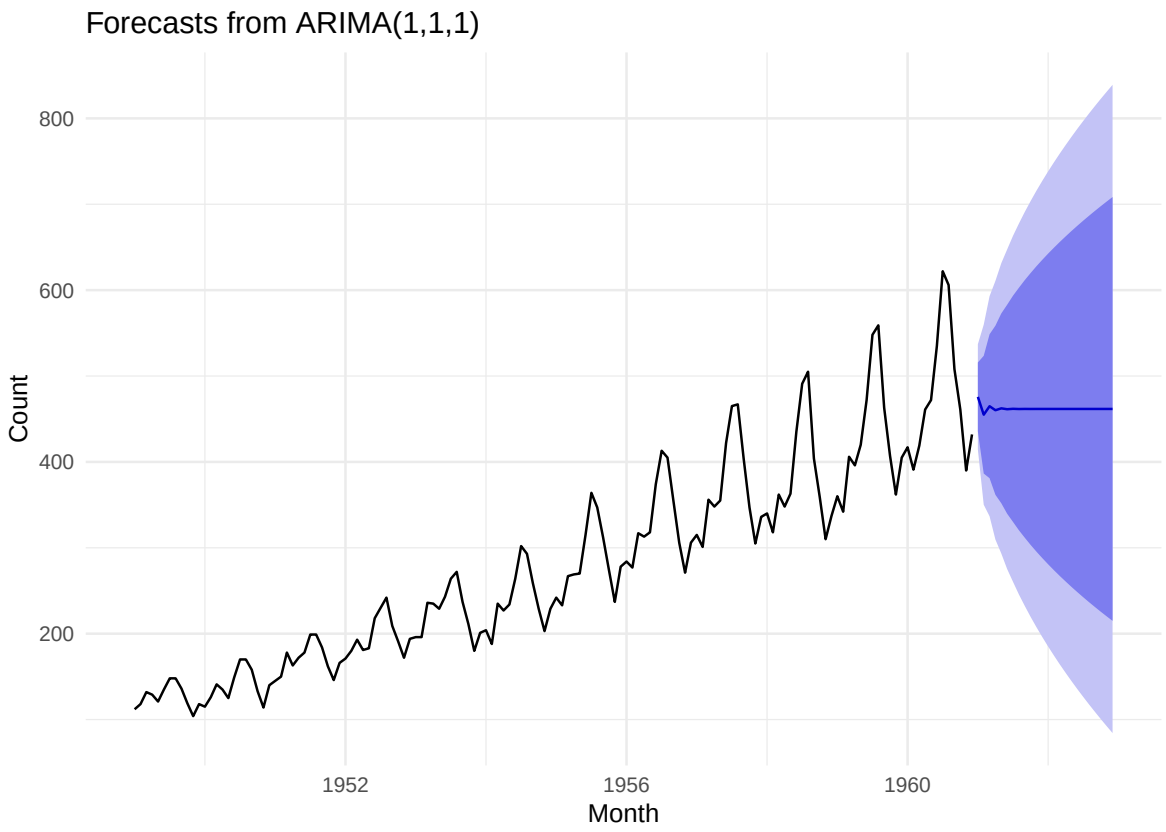


Figure 9: An ARIMA(1,1,1) model fitted to `AirPassengers` series with forecasts and prediction intervals for 24 months.

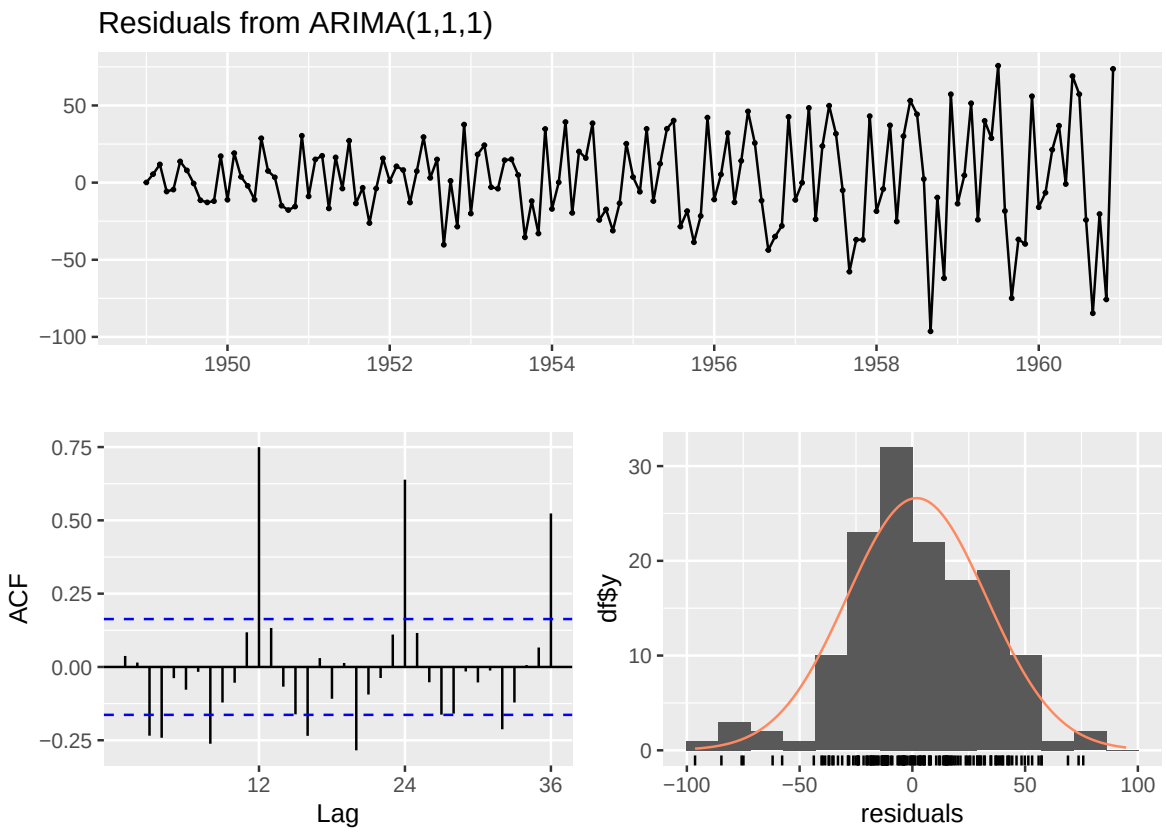


Figure 10: Residual diagnostics for the fitted ARIMA(1,1,1) model: the residual series, its ACF and a histogram against a normal density.

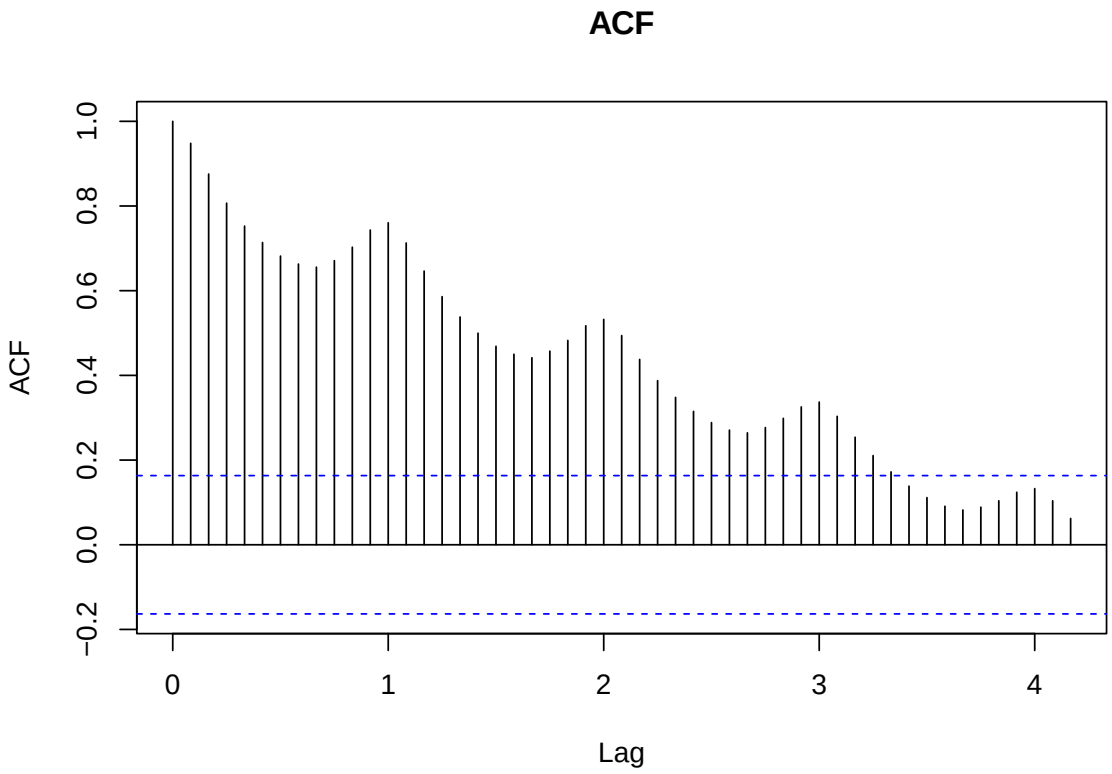


Figure 11: ACF plot of AirPassengers series to illustrate seasonality.

```
# Which SARIMA model is the best fit:
best_fit = auto.arima(AirPassengers,
                      lambda=0,
                      stepwise=FALSE,
                      approximation=FALSE)
summary(best_fit)
```

```
Series: AirPassengers
ARIMA(0,1,1)(0,1,1)[12]
Box Cox transformation: lambda= 0

Coefficients:
          ma1     sma1
      -0.4018  -0.5569
s.e.   0.0896   0.0731

sigma^2 = 0.001371:  log likelihood = 244.7
AIC=-483.4   AICc=-483.21   BIC=-474.77

Training set error measures:
                     ME     RMSE      MAE           MPE     MAPE     MASE
Training set 0.05140418 10.15504 7.357553 -0.004079016 2.623636 0.229706
                    ACF1
Training set -0.03689673
```

Figure 12 carries the seasonal shape forward while the intervals widen. This is the behaviour a seasonal model is fitted for.

Conditional Variance Models

Spectral Analysis

Section 2

Data Statement

[cite: 1]

Teaching Session Summaries

[cite: 1]

Worked Tutorials & Software Implementations

[cite: 1]

Topic Exploration

[cite: 1]

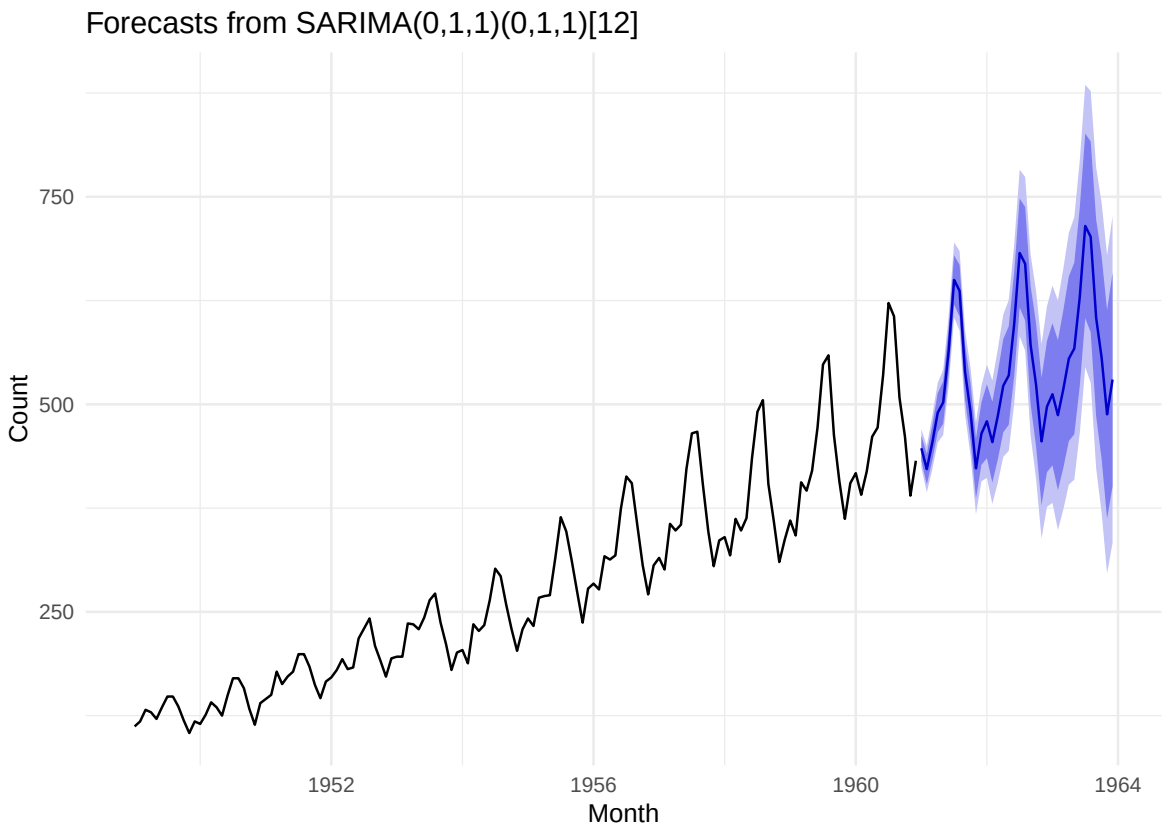


Figure 12: A SARIMA(1,1,1)(0,1,1) model fitted to AirPassengers series with a stochastic level and an evolving seasonal pattern, forecast three years ahead.

Projects

Overview

This section contains four substantial projects spanning Section 1 and Section 2[cite: 1].

Project	Section	Project Type	Focus Area	Link
Project 1	Section 1	Simulation Study	Model Comparison	Summary[cite: 1]
Project 2	Section 1	R Package Implementation	Computational Programming	Summary[cite: 1]
Project 3	Section 2	Shiny Application	Interactive Visualization	Summary[cite: 1]
Project 4	Section 2	Exploratory Data Analysis	Real-World Data	Summary[cite: 1]

Project 1

This is just basic EDA so far. I might only use this dataset to fit a hidden markov model once we've covered that topic in class.

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

```
#|echo: false
#|warning: false
# Load required libraries:
library(readr)
library(tidyverse)
library(tsibble)
```

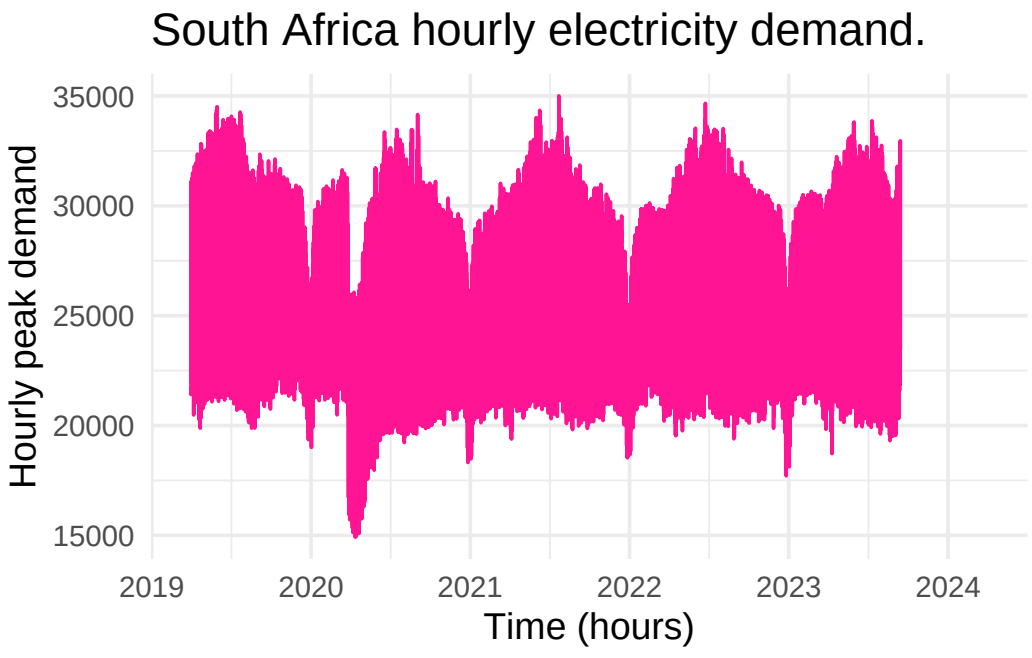
```r
library(lubridate)
library(dplyr)
```

```r
# Read in dataset:
dat = read_csv('_data/ESK6816.csv')
# View(dat)
```

```r
# Convert data to time series object:
demand_ts = ts(
  dat$`RSA Contracted Demand`,
  start = c(2019, 2161), # Hour 2161 of 2019 (April 1st)
  frequency = 8766       # Hourly frequency per year
)

# Convert time series object to dataframe (for plotting):
demand_df = data.frame(
  hour    = as.numeric(time(demand_ts)),
  count   = as.numeric(demand_ts)
)

# Plot:
ggplot(data = demand_df,
       mapping = aes(x = hour,
                     y = count)) +
  geom_line(col = 'deeppink')  +
  theme_minimal(base_size=14) +
  labs(title = 'South Africa hourly electricity demand.',
       x = 'Time (hours)',
       y = 'Hourly peak demand')
```



As seen in the above plot, we cannot get a clear idea of the seasonality. We need to

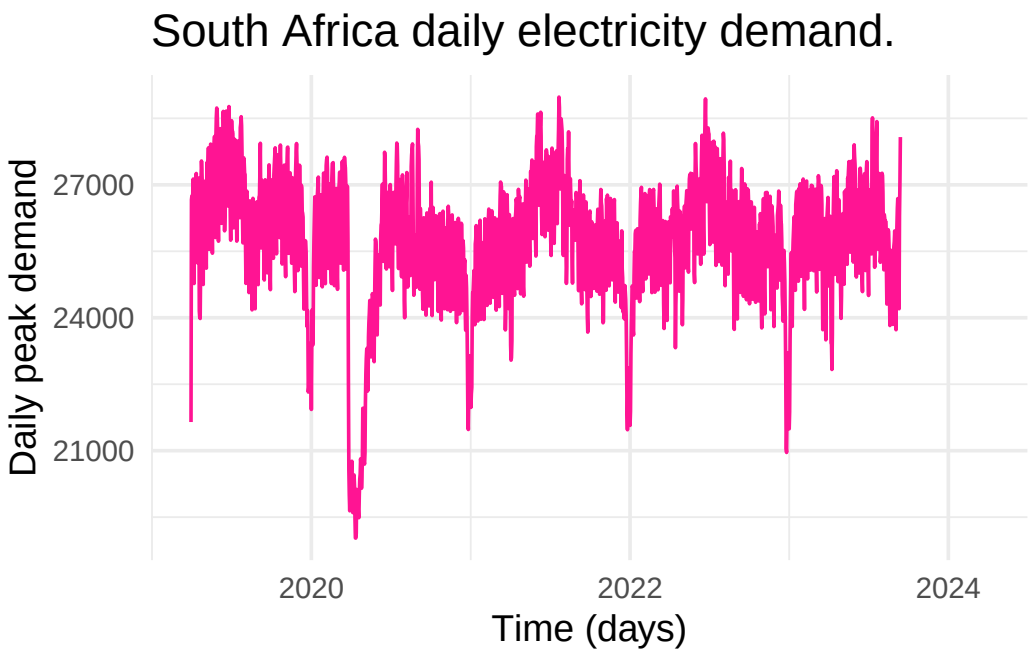
aggregate the data.

To handle the leap years present in the data we will use tsibble.

```
# Create a tsibble dataframe:
df_ts = tibble(datetime=seq(from=as.POSIXct('2019-04-01 00:00:00'),
                            length.out=length(demand_ts),
                            by='hour'),
               demand=as.numeric(demand_ts)
               ) |>
  as_tsibble(index=datetime)

# Aggregate to daily values:
demand_daily_ts = df_ts |>
  index_by(date=~as.Date(.))|>
  summarise(demand_mean=mean(demand,na.rm=TRUE))

# Plot:
ggplot(data = demand_daily_ts,
       mapping = aes(x = date,
                     y = demand_mean)) +
  geom_line(col = 'deeppink')  +
  theme_minimal(base_size=14) +
  labs(title = 'South Africa daily electricity demand.',
       x = 'Time (days)',
       y = 'Daily peak demand')
```

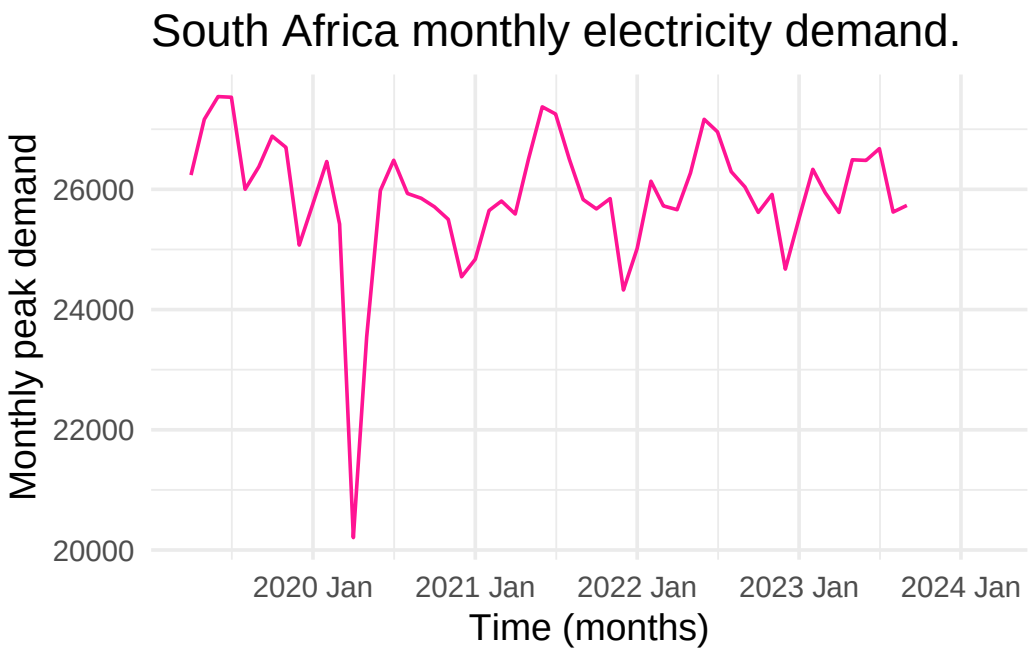


Again, there is still a lot of unnecessary noise in the data so we're going to aggregate to months.

```
# Aggregate to monthly values:
```

```r
demand_monthly_ts = df_ts |>
  index_by(year_month=~yearmonth(.))|>
  summarise(demand_mean=mean(demand,na.rm=TRUE))

# Plot:
ggplot(data = demand_monthly_ts,
       mapping = aes(x = year_month,
                     y = demand_mean)) +
  geom_line(col = 'deeppink')  +
  theme_minimal(base_size=14) +
  labs(title = 'South Africa monthly electricity demand.',
       x = 'Time (months)',
       y = 'Monthly peak demand')
```



Project 2

The plan for this project is to fit all the autoregressive models to both datasets, compare them and forecast using the best models. I'm hoping to then use these same datasets in another projects to see if SST and historic dam levels can be used to predict drought in Southern Africa (I'm not 100% sure how I'm going to do this but I think it'll be interesting)

Purely for interests sake, here is a website that maps the sea surface temperature anomlaies: https://earth.nullschool.net/#current/ocean/surface/currents/overlay=sea_surface_temp_anom 100.92,9.72,366/loc=-99.041,-1.152

SST along the equatorial band in the eastern and central Pacific is much higher than past temperatures. This is aligning with experts prediciting a super El Nino in 2026/2027.

- **Section & Topic:** Autoregressive Models
- **Project Type:** Model implementaion
- **Question / Aim:** Is sea surface temperature in the eastern Pacific Ocean and Theewaterskloof dam levels predictable?

- **Data Used:** [https://www.pmel.noaa.gov/tao/drupal/disdel/] [https://odp-cctegis.opendata.arcgis.com/] I will reference this properly!
- **Methods Used:** AR,MA,ARMA,ARIMA,SARIMA,SARIMAX (at least this is the plan)

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Predicting drought in Southern Africa

The aim of this project is to predict Theewaterskloof dam levels as well changes in sea surface temperatures in the eastern Pacific Ocean (along 95W) and use this data to predict drought occurrence in Southern Africa.

The El Nino phase of the El Nino Southern Oscillation (ENSO) is characterise by anomalous warming of sea surface temperatures in the eastern and central Pacific Ocean. This phenomenon typically starts between March and June, reaching its peak in December. An El Nino event causes warm and dry conditions over Southern Africa, typically taking effect 3-6 months after the onset of El Nino, in the Pacific. Over the 2015-2016 season, the El Nino weather patterns resulted in prolonged drought in Southern Africa with dam levels dropping and severe water restrictions being implemented.

The South African Weather Service (SAWS) is warning against a super El nino event in 2026-2027, as current sea surface temperatures rise beyond the threshold used to define a very strong El Nino.

We are interested in the Theewaterskloof dam current levels as Theewaterskloof is the main supplier of water to Cape Town.

References:

[https://onlinelibrary.wiley.com/doi/abs/10.1002/env.476?casa_token=Eg_zhVj43usAAAAA%3Ar0W
l0YIV8mbMZnQvBcHiz9Cc_yBXPmfHgftiHsExL1riAtWaHZKw]

[https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(03)12421-X/fulltext]

[https://mg.co.za/the-green-guardian/2026-09-01-the-saws-warns-exceptionally-strong-2026-2027-el-nino-could-bring-severe-heatwaves-to-sa/]

Exploratory Data Analysis

In the `Dam_Levels_from_2000` dataset there was one duplicated observation. The second observation was removed as, compared to the other observations, it appeared to be a mistake.

I need better justification for this

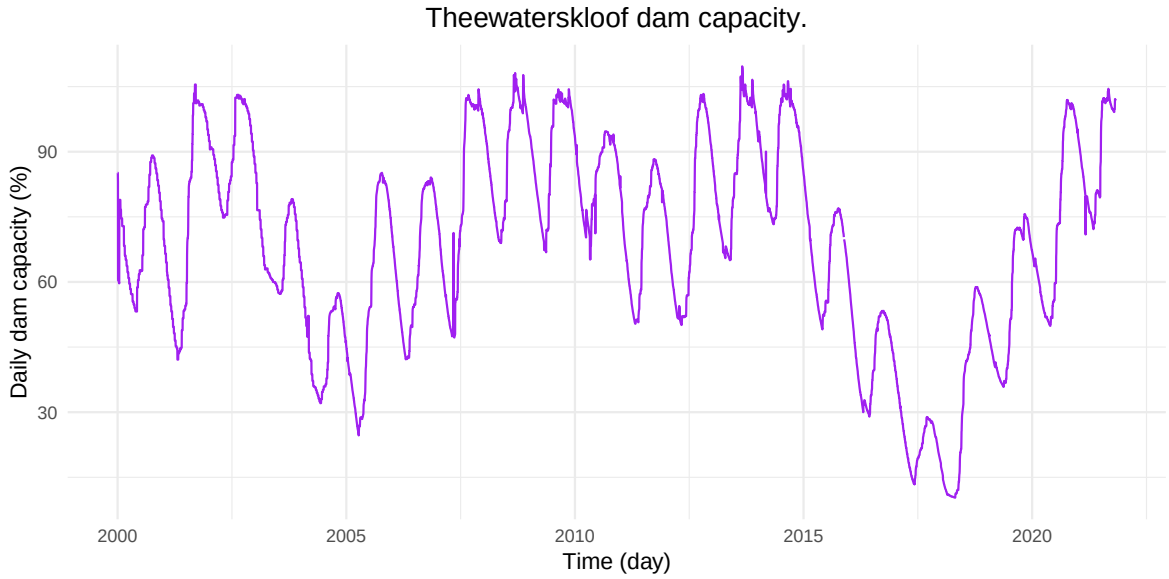


Figure 13: Theewaterskloof dam capacity from January 2000 to October 2021.

As seen in both Figure 13 and Figure 14 there is noise in the daily observations. The data is aggregated to monthyl observations to obtain a clearer visualisation of the seasonality present in the data.

From Figure 15 and Figure 16 it is apparent that there is seasonality in the dataset.

We have 21 years worth of data. The `Dam_Levels_from_2000` dataset has observations until January 2026 but due to missing observations in the `sst` dataset we have truncated the data. We are interested in forecasting until 2028. As the forecast horizon grows we can expect increased uncertainty in our predictions. Experts are prediciting an exceptionally strong El Nino event in 2026/2027. It will be interesting to see if these trends are picked up.

Southern Africa experienced severe droughts in 2015/2016 and 2018/2019. These years also saw significant El nino conditions.

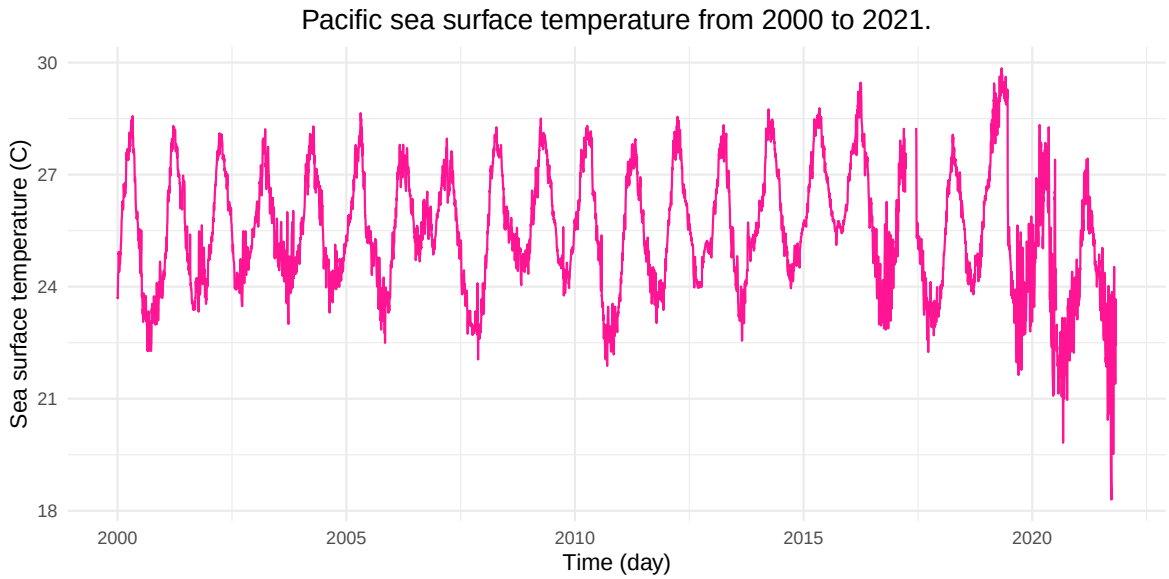


Figure 14: Average Pacific sea surface temperature from January 2000 to October 2021.

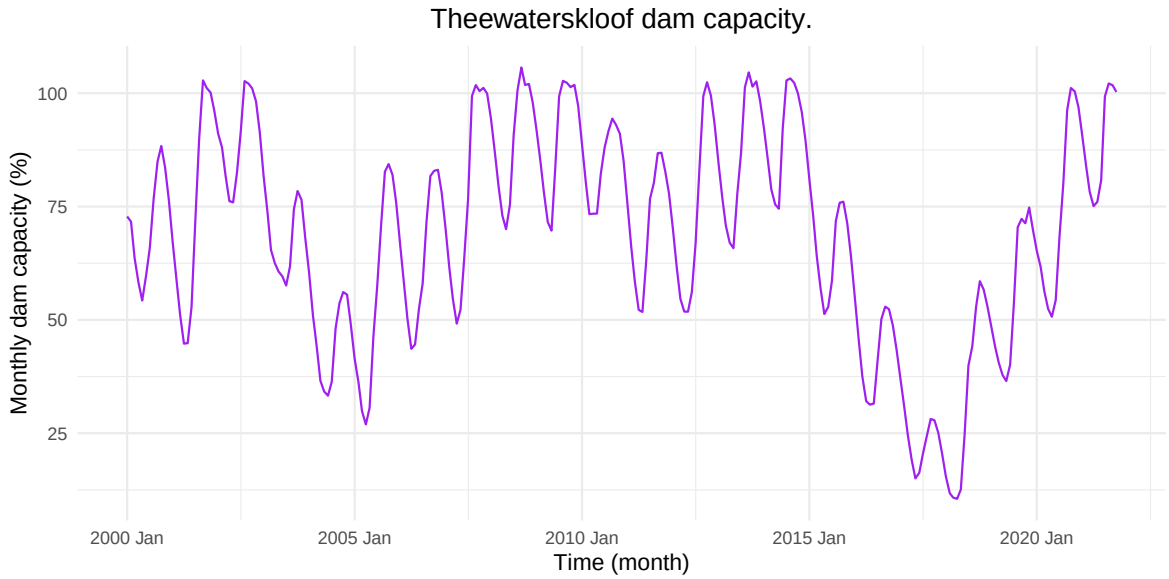


Figure 15: Theewaterskloof dam capacity from January 2000 to October 2021.

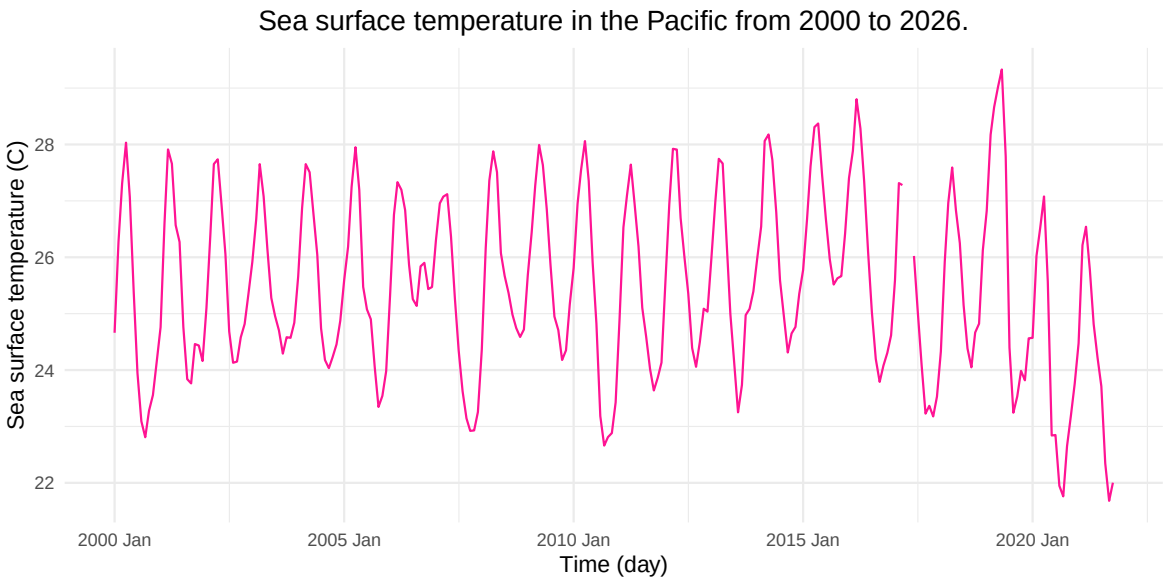


Figure 16: Average Pacific sea surface temperature from January 2000 to October 2021.

I need to have a closer look at the data, the sst data has observations all the way up until today (continuoulsy uodating). It's also worth noting that there is approximately a 3-6 month lag between increased sst and drought in southern africa - we could expand the number of observations for the dam levels dataset by approximately 6 months because of this

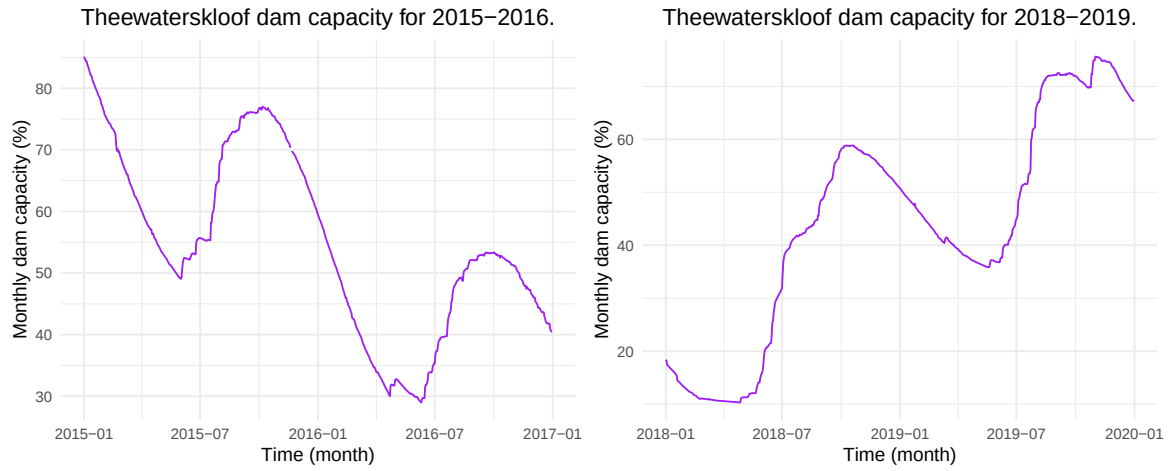


Figure 17: Theewaterskloof dam capacity fro 2015-2016 and 2018-2019.

Look at the actual El Nino events here as well as sea surface temperatures 6 months before these observations for comparison.

The sea surface temperatures and the dam levels have inverse trends. When sea surface tempertures are high, dam levels are low.

Need to mention autocorrelation herte.

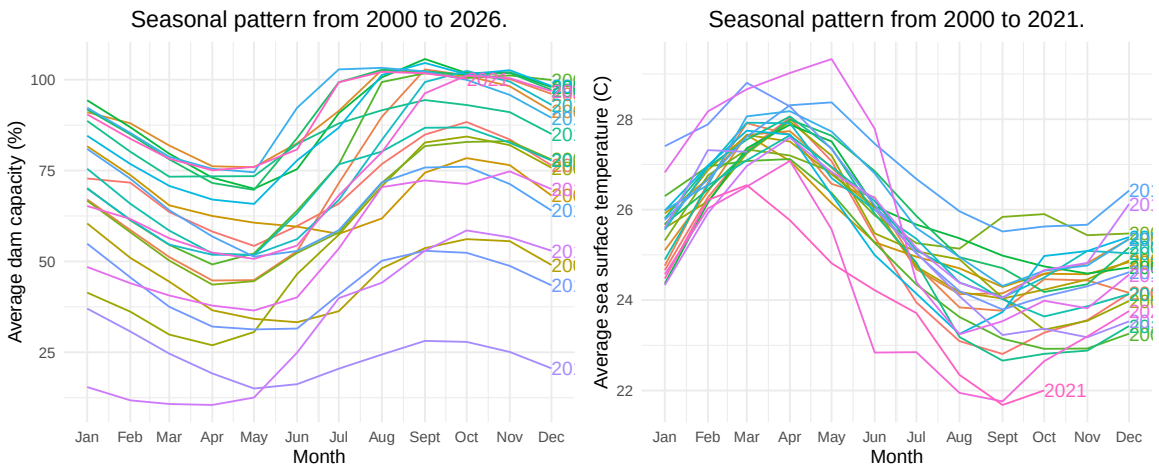


Figure 18: Seasonal pattern of Theewaterskloof dam capacity and average Pacific sea surface temperature.

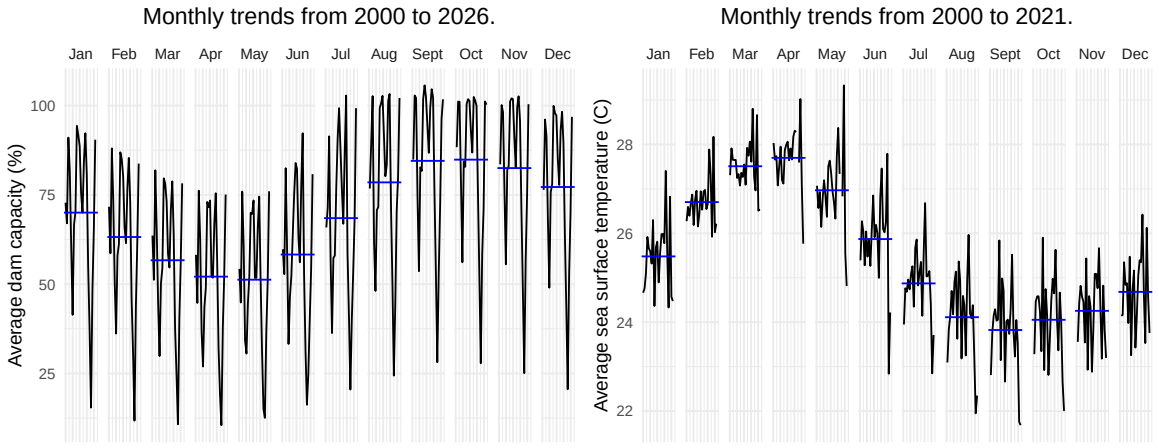


Figure 19: Monthly trends in Theewaterskloof dam capacity and average Pacific sea surface temperature.

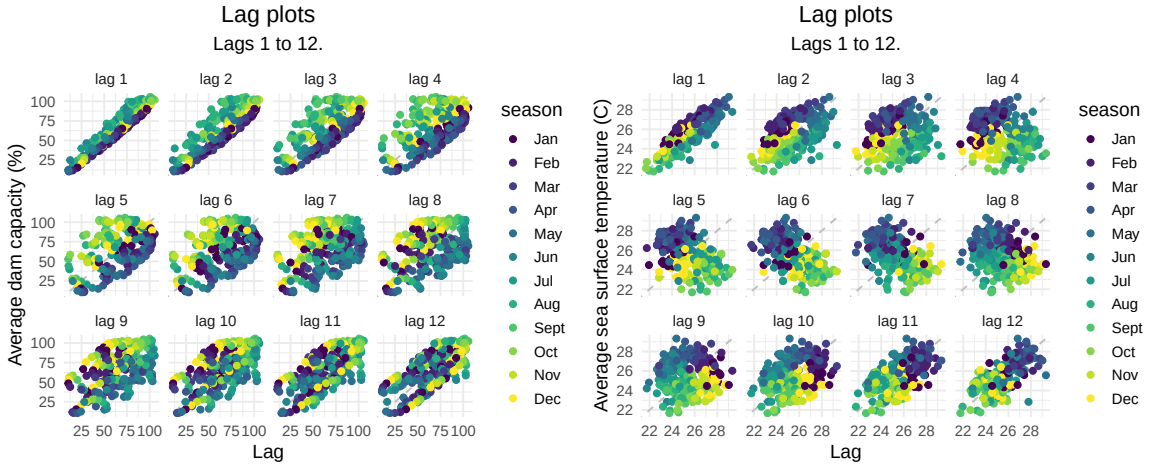


Figure 20: Lad plots of Theewaterskloof dam capacity and average Pacific sea surface temperature.

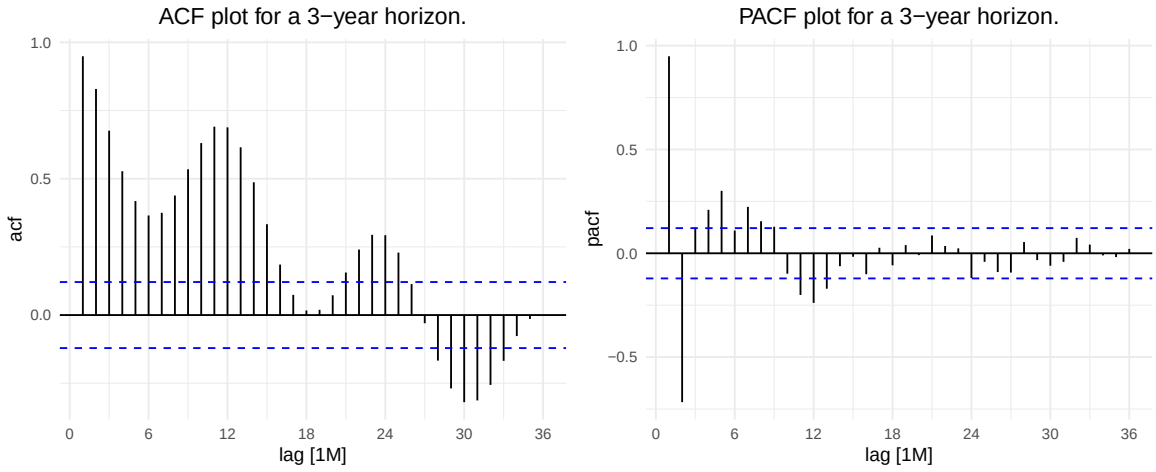


Figure 21: ACF and PACF plots for Theewaterskloof dam capacity.

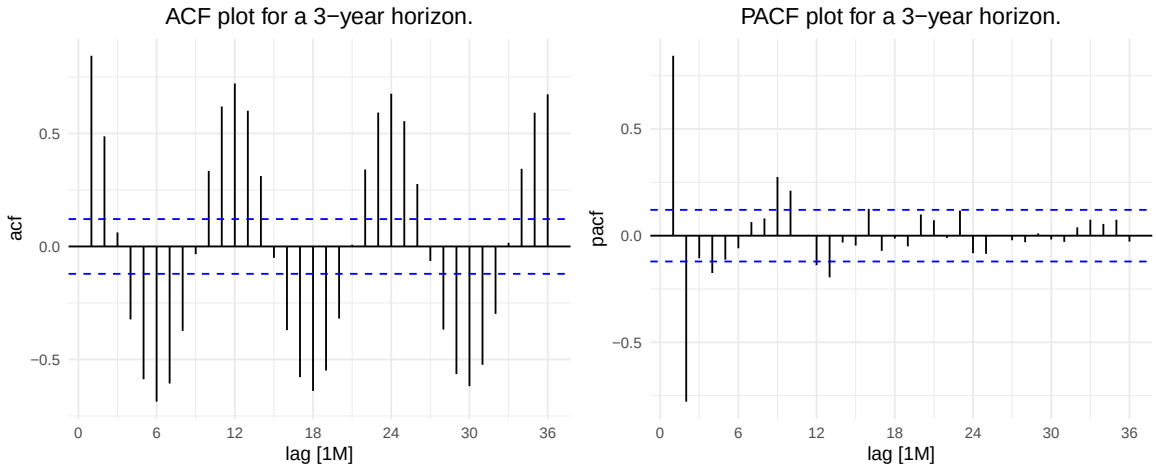


Figure 22: ACF and PACF plots for Pacific sea surface temperature.

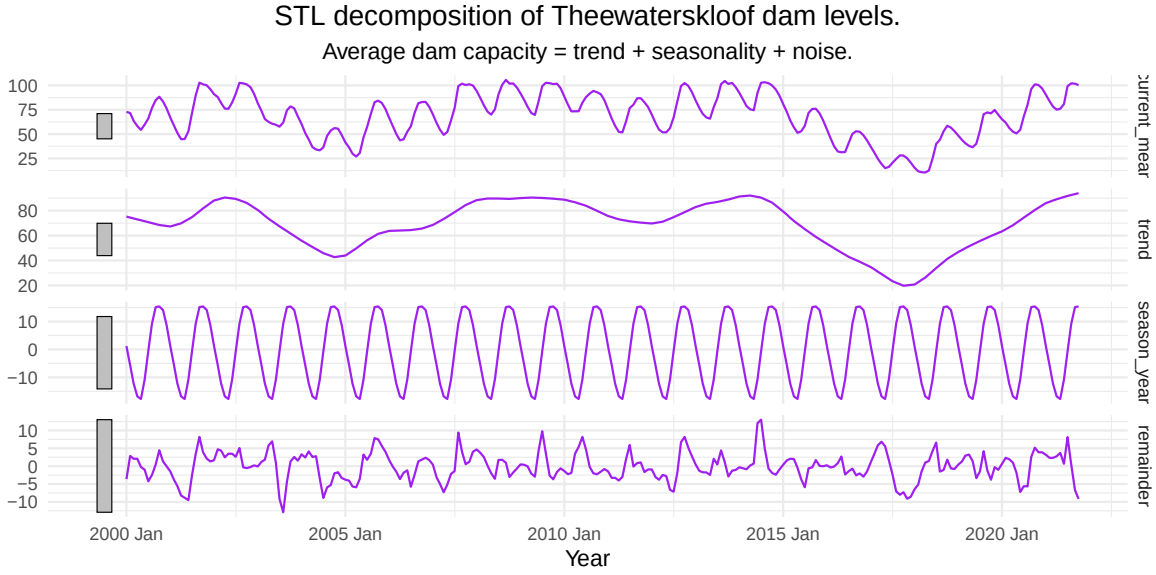


Figure 23: Decomposition of Theewaterskloof dam capacity series.

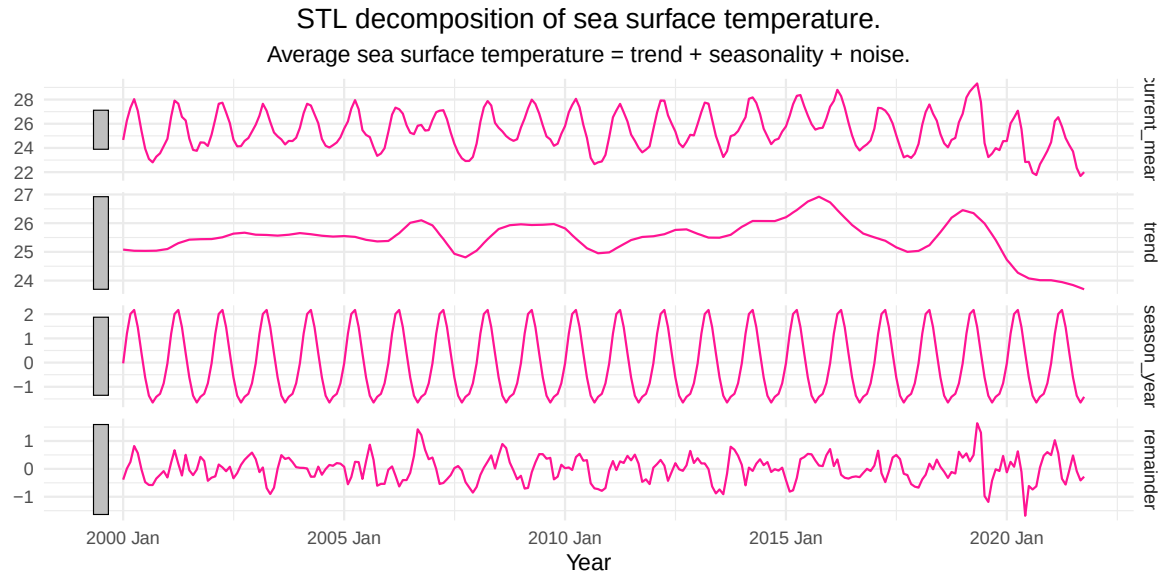


Figure 24: Decomposition of average Pacific sea surface temperature series.

Comment on trends, seasonality etc for both sst and dam levels.

```
        Mean    Naive Seasonal naive    Drift
[1,] 26.1521 7.320172       17.53497 7.320172
```

Here we can see the errors assocated with each of our benchmark models. Since the Naive and Drift models both have the same error associated with them we will use the Drift model as a comparison to the other time series models used.

```
# A tibble: 1 x 1
  nsdiffs
    <int>
1       1
```

```
# A tibble: 1 x 1
  ndiffs
   <int>
1      0
```

```
# A tibble: 1 x 1
  nsdiffs
    <int>
1       1
```

```
# A tibble: 1 x 1
  ndiffs
   <int>
1      0
```

We only need to seasonally difference the `Dam_Levels_from_2000` and `sst` datasets, no linear differencing is required.

Model fitting

I have naively fitted MA,AR,ARMA,ARIMA,SARIMA here just to get a gauge of how they look.

Dam levels

```
# A mable: 6 x 2
# Key:     Model [6]
  Model                                  Fit
  <chr>                                <model>
1 ar               <ARIMA(1,0,0) w/ mean>
2 ma               <ARIMA(0,0,1) w/ mean>
3 arma             <ARIMA(1,0,1) w/ mean>
4 arima                    <ARIMA(1,1,1)>
5 sarima       <ARIMA(1,0,1)(1,1,1)[12]>
6 auto_sarima  <ARIMA(3,0,1)(1,1,0)[12]>
```

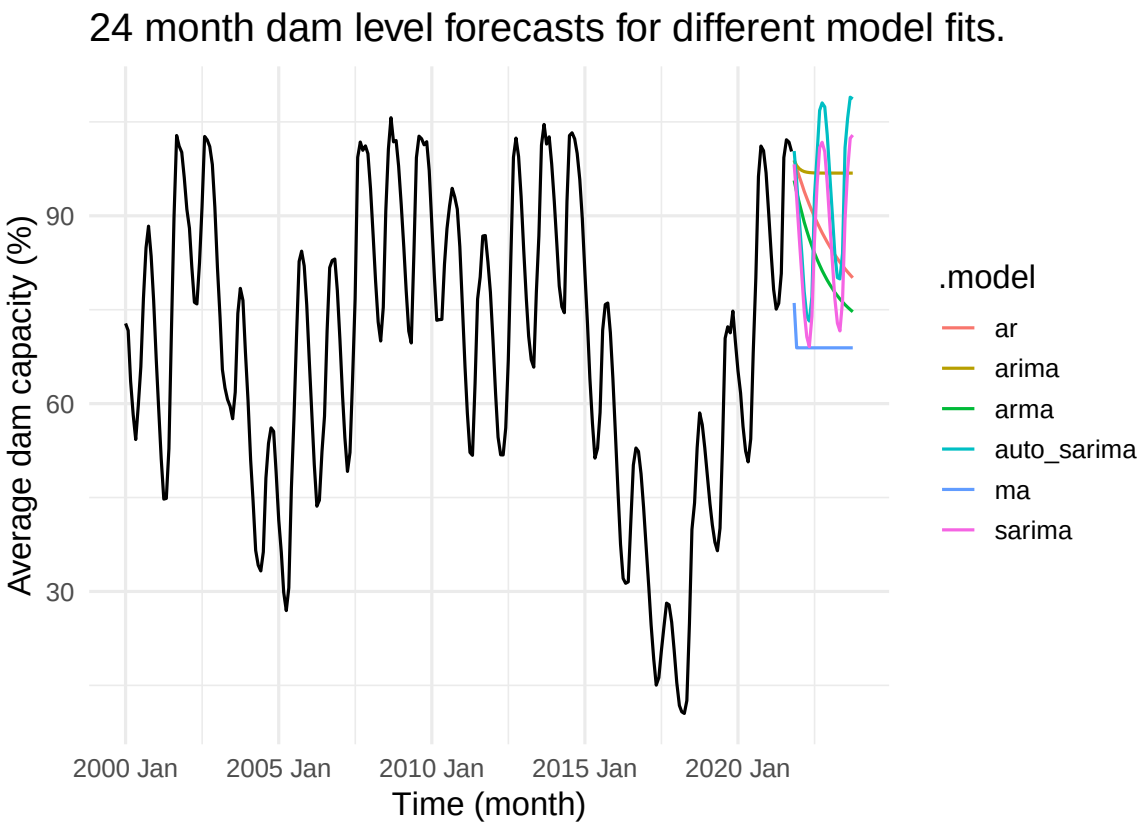


Figure 25: Naive forecasting of fitted models.

Sea surface temperature

```
# A mable: 6 x 2
# Key:     Model [6]
  Model                                  Fit
```

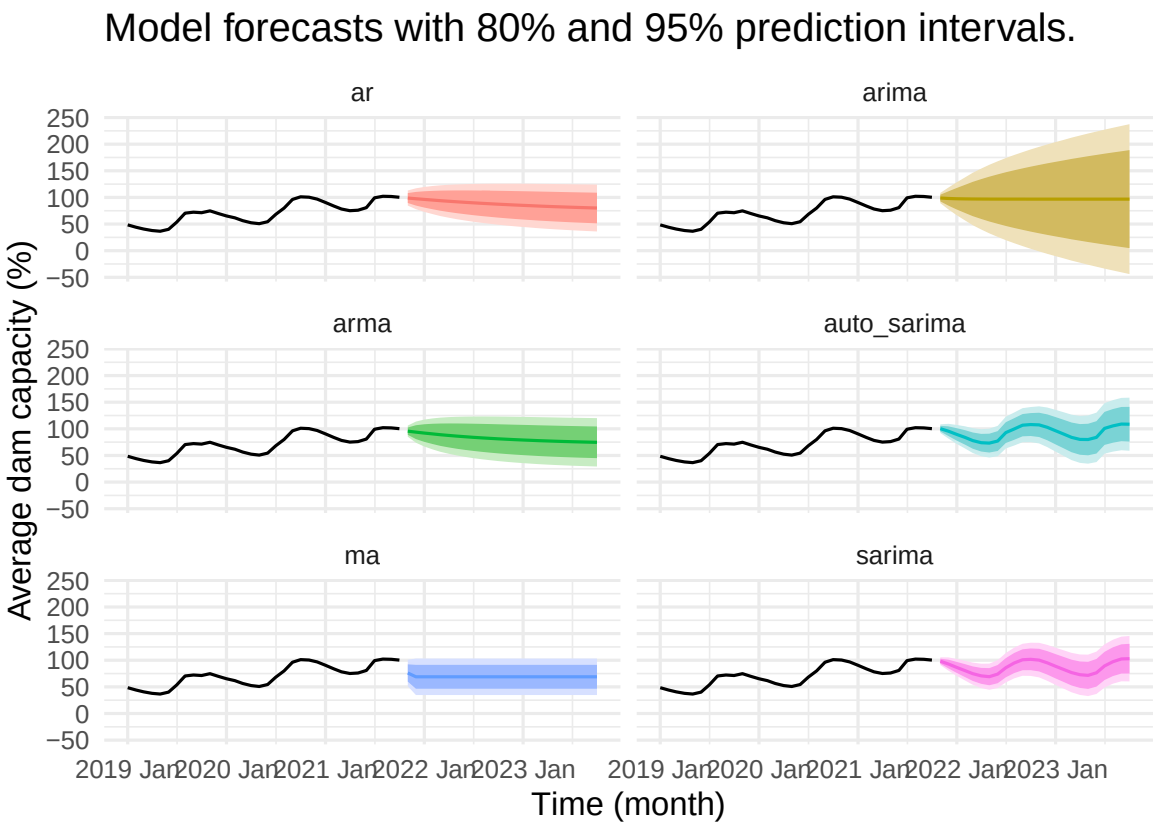


Figure 26: Naive forecasting of fitted models.

```
  <chr>                            <model>
1 ar                 <ARIMA(1,0,0) w/ mean>
2 ma                 <ARIMA(0,0,1) w/ mean>
3 arma               <ARIMA(1,0,1) w/ mean>
4 arima                        <ARIMA(1,1,1)>
5 sarima          <ARIMA(1,0,1)(1,1,1)[12]>
6 auto_sarima     <ARIMA(2,0,1)(2,1,0)[12]>
```

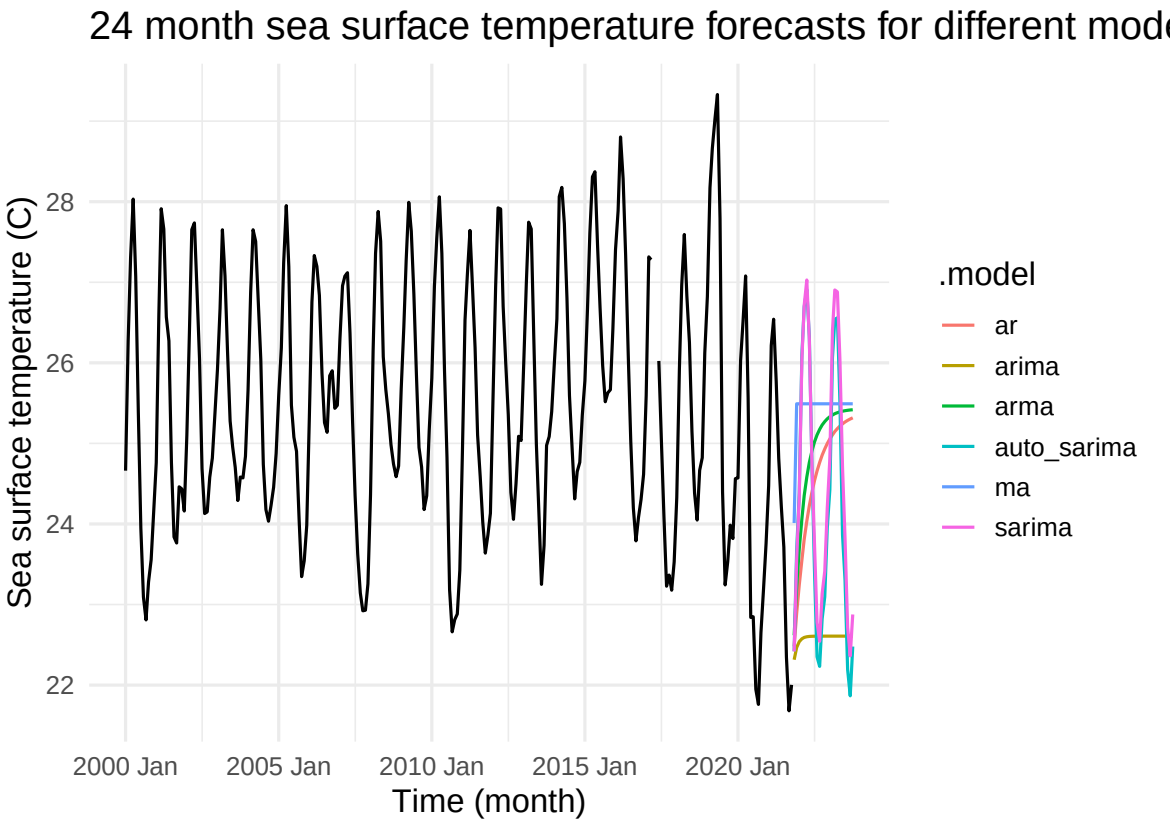


Figure 27: Naive forecasting of fitted models.

Project 3

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

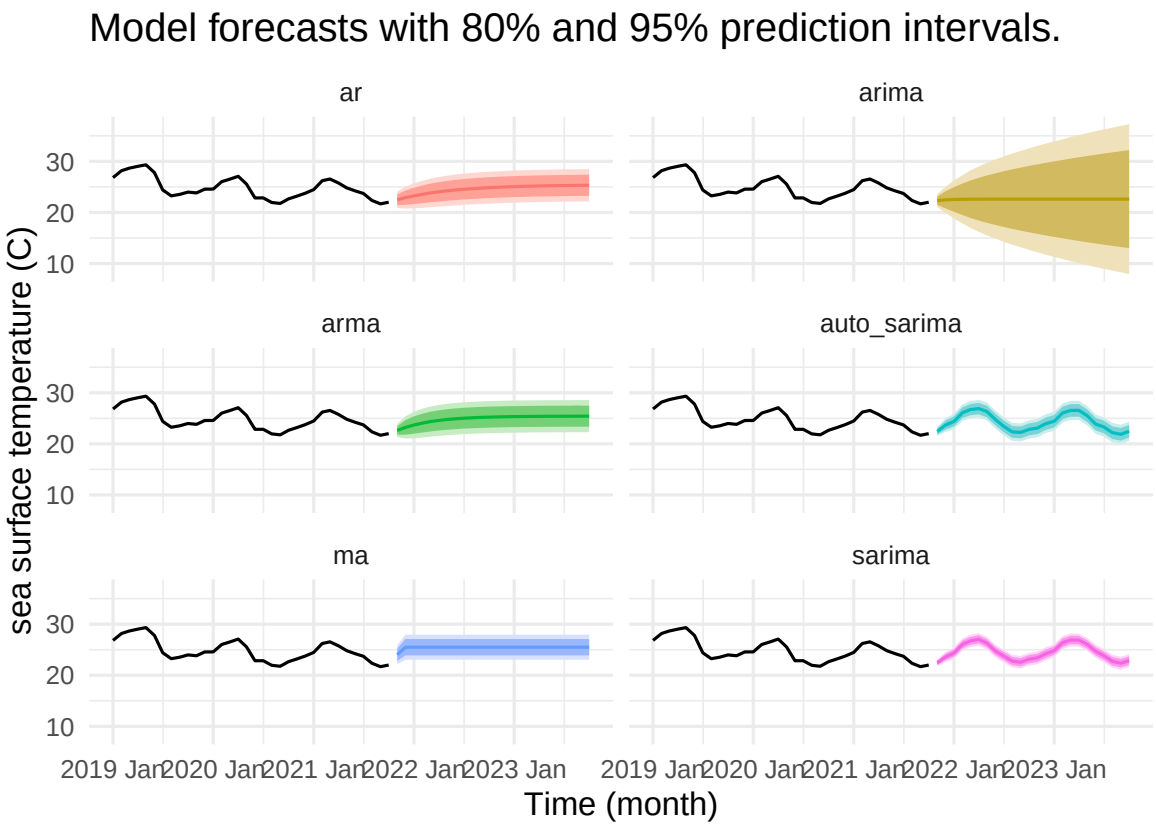


Figure 28: Naive forecasting of fitted models.

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Project 4

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- Detailed Analysis & Code Script[cite: 1]

Journal

This journal documents my continuous learning progress throughout the course[cite: 1].

- View Chronological Entries[cite: 1]

Appendix

R Code & Scripts

- `scripts/simulation-study.R`
- `scripts/shiny-app/app.R`

Data Access & Repositories

- Raw Data Files & Data Access Instructions[cite: 1]

Computational Notebooks

- Extended Analysis Reports[cite: 1]